Universitatea POLITEHNICA din București
Facultatea de Automatică și Calculatoare

Electronică analogică

Curs – Modulul 9

Circuite fundamentale cu amplificatoare operaționale

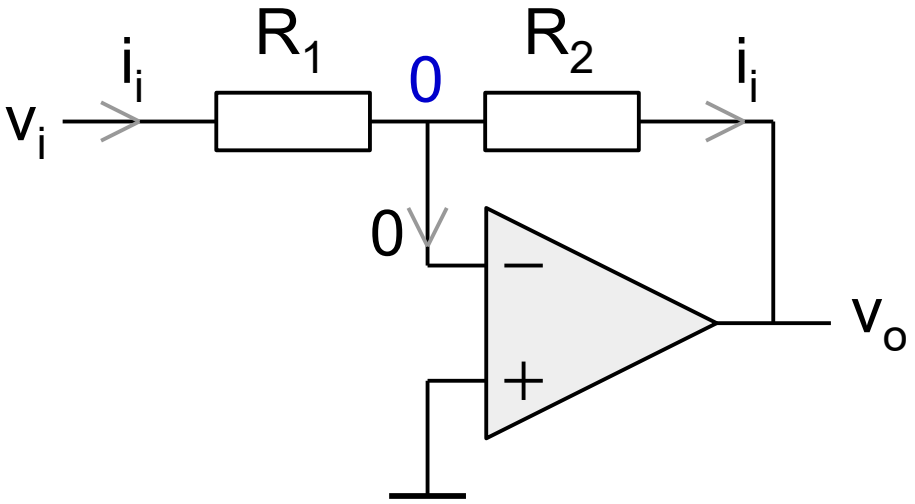
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2022-2023

Sumar

- Amplificatoare:
  - cu o singură intrare
    - **neinversor**
      - **repetor**
    - **inversor**
  - cu multiple intrări
    - **sumator**
    - **de diferență**

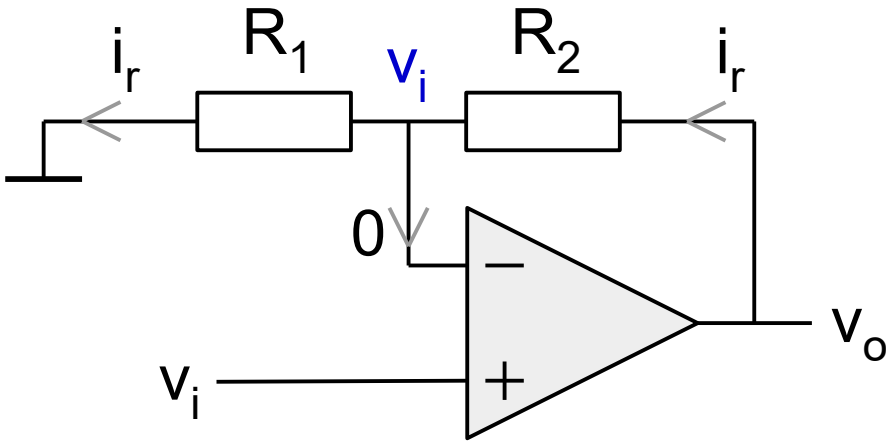
Ne amintim: AO ideale



$$v_- = 0 \qquad i_i = \frac{v_i}{R_1} = -\frac{v_o}{R_2}$$

$$v_o = -\frac{R_2}{R_1} v_i$$

amplificator **inversor**

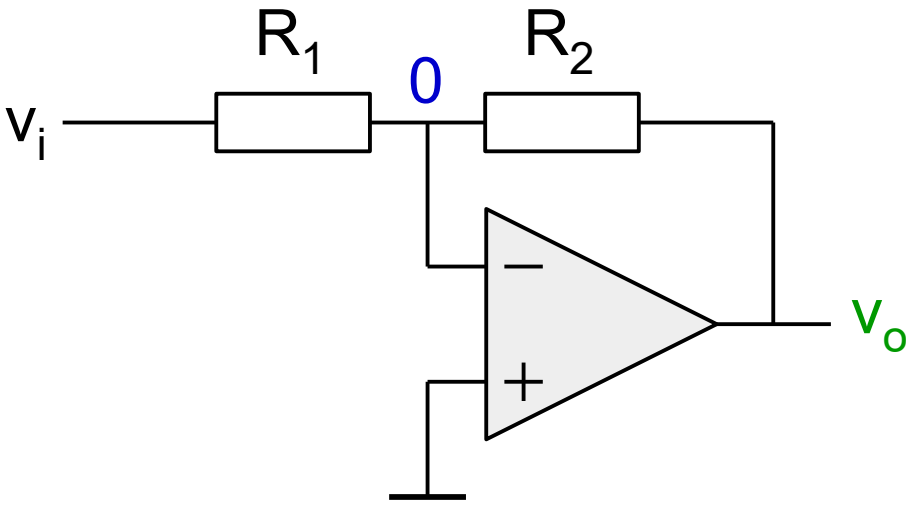


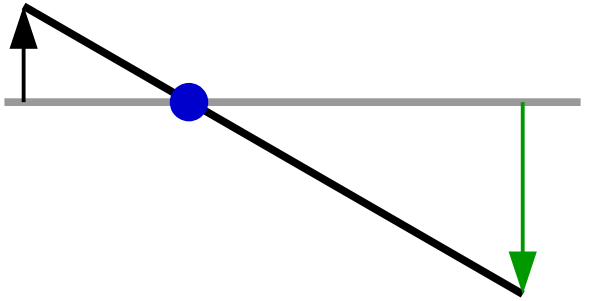
$$v_- = v_i \qquad i_r = \frac{v_i}{R_1} = \frac{v_o}{R_1 + R_2}$$

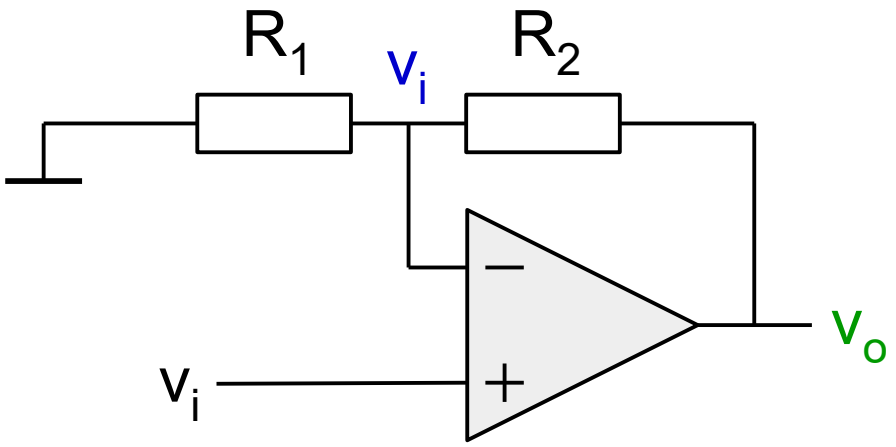
$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_i$$

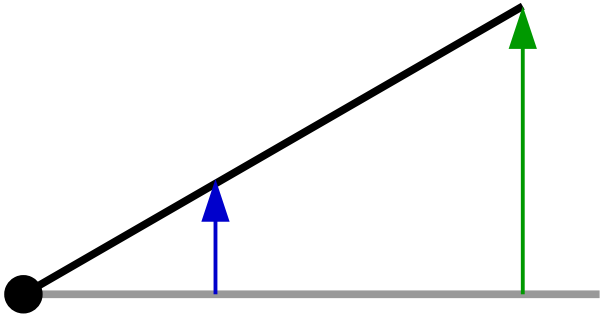
amplificator **neinversor**

Analogie mecanică

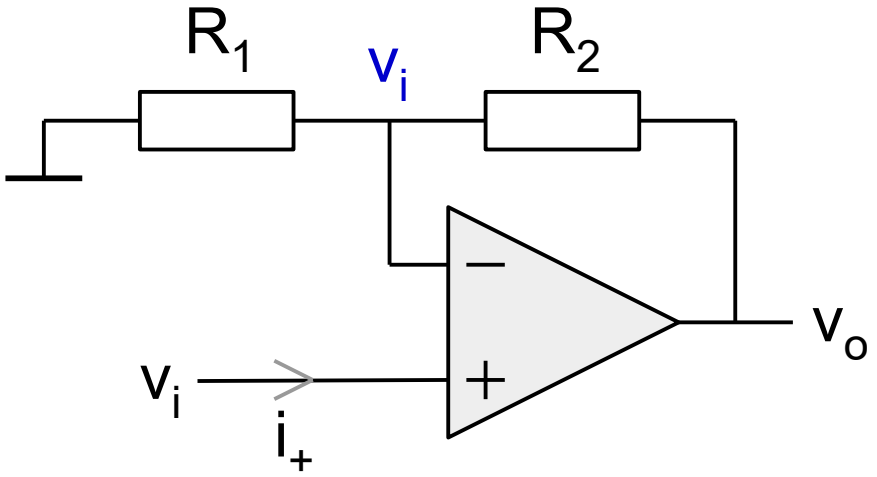








Amplificatorul neinversor (1)



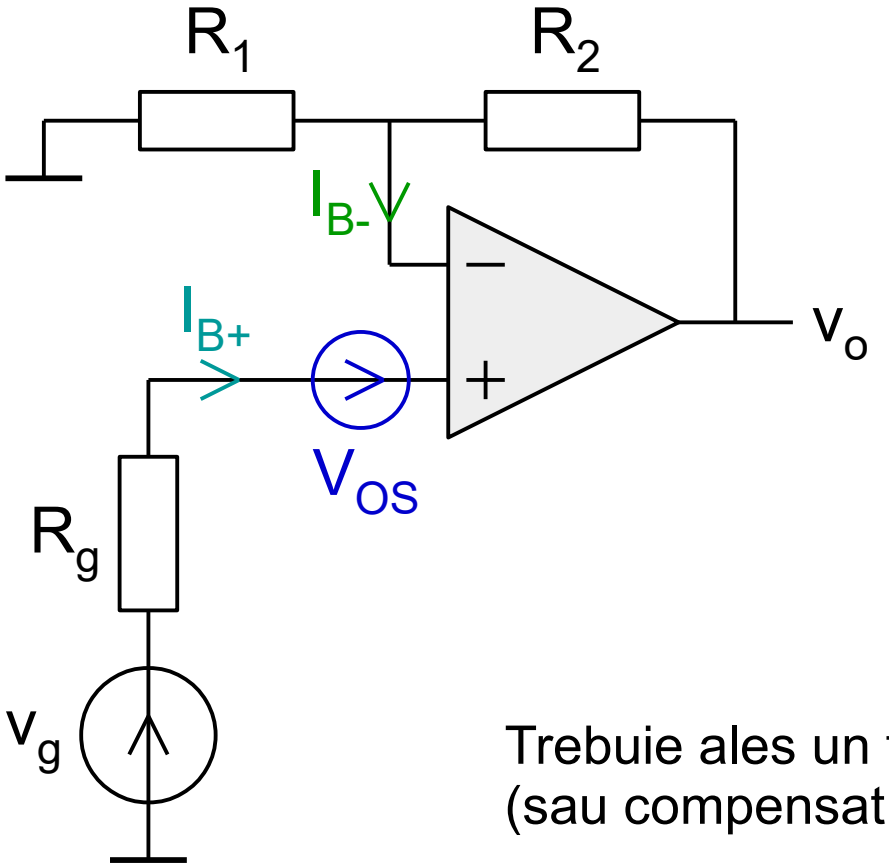
ideal:

$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_i$$

$$i_+ = 0$$

Important:
Neinversorul prezintă o **impedanță de intrare foarte mare**
– nu afectează semnificativ sursa de semnal

Amplificatorul neinversor (2)

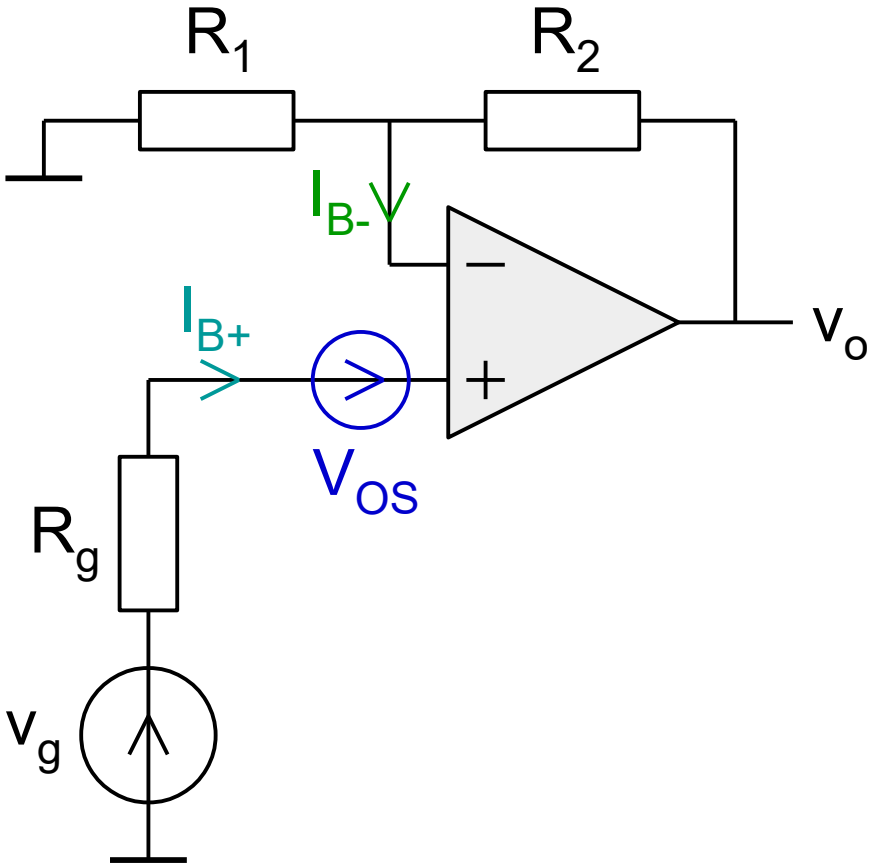


cu neidealități:
tensiune de decalaj (*offset voltage*) V_{OS}
curenți de polarizare (*bias currents*) I_{B+}, I_{B-}

Trebuie ales un tip de AO având $V_{OS} \ll v_g$
(sau compensat efectul)

$$v_o = \left(1+\frac{R_2}{R_1}\right) v_g + \left(1+\frac{R_2}{R_1}\right) V_{OS} - I_{B+} R_g \left(1+\frac{R_2}{R_1}\right) + I_{B-} R_2$$

Amplificatorul neinversor (3)



dacă este posibil se alege

$$R_g = R_1 \| R_2$$

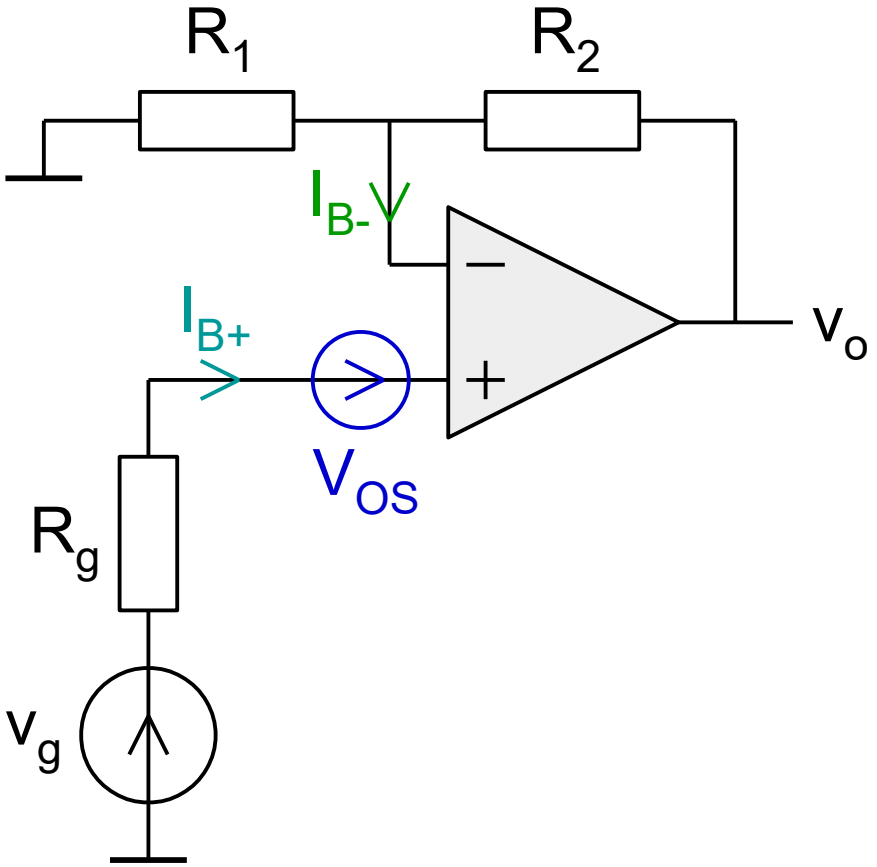
OBS: are sens doar dacă AO folosit are

$$I_{OS} \ll I_B \quad \left(I_{B+} \approx I_{B-}\right)$$

curent de decalaj (*offset current*) I_{OS}

$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_g + \left(1 + \frac{R_2}{R_1}\right) V_{OS} + \overbrace{\left(I_{B+} - I_{B-}\right)}\, R_2$$

Amplificatorul neinversor (4)



altfel:
se aleg tipul de AO și rezistențele a.î.:

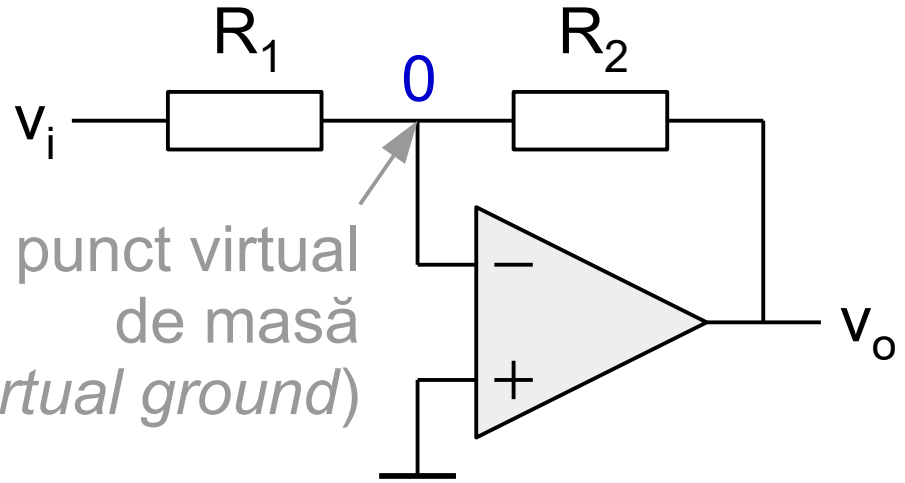
$$V_{OS} \ll v_g$$

$$I_B R_g \ll v_g$$

$$I_B R_1 \ll v_g$$ (de ce R_1 nu R_2?)

$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_g + \left(1 + \frac{R_2}{R_1}\right) V_{OS} - I_{B+} R_g \left(1 + \frac{R_2}{R_1}\right) + I_{B-} R_2$$

Amplificatorul inversor (1)



ideal:

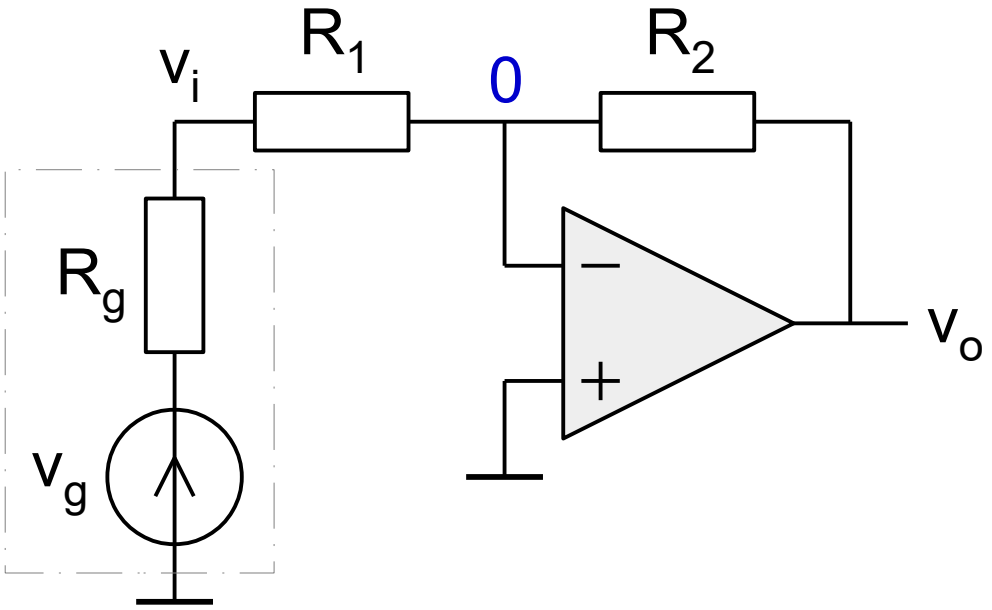
$$v_o = -\frac{R_2}{R_1} v_i$$

$$i_i = \frac{v_i}{R_1} \qquad Z_i = R_1$$

Important:
Inversorul prezintă o **impedanță de intrare egală cu R_1**

– sursa de semnal trebuie să aibă **impedanță mică**
(ex.: ieșirea din alt AO)

Amplificatorul inversor (2)



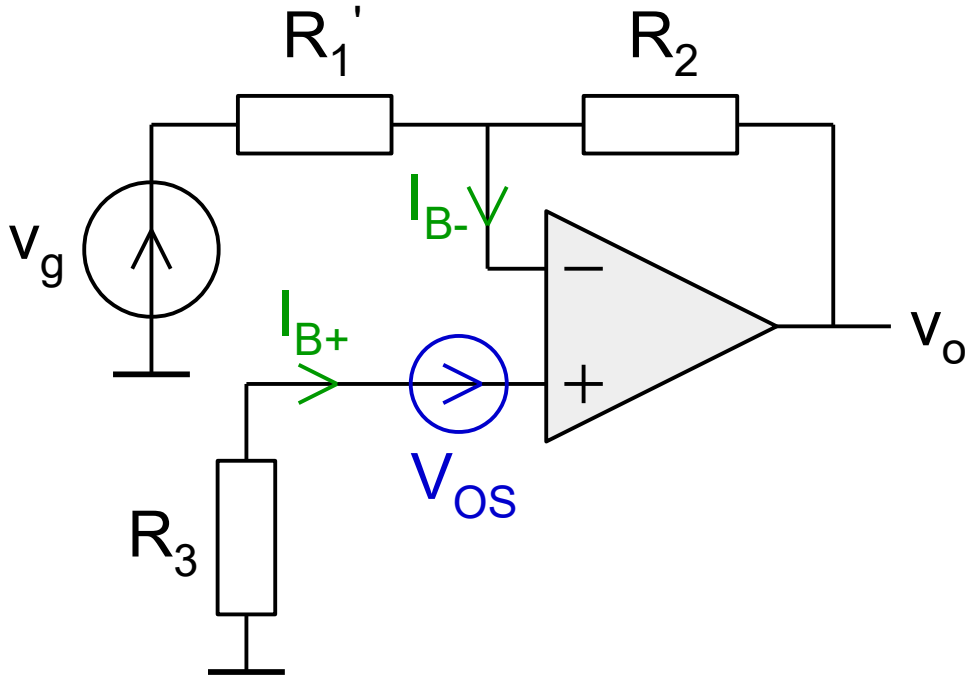
$$v_i = v_g \frac{R_1}{R_1 + R_g}$$

efectul R_g:

$$v_o = -\frac{R_2}{R_1} v_i$$

$$v_o = -\frac{R_2}{R_1 + R_g} v_g$$

Amplificatorul inversor (3)



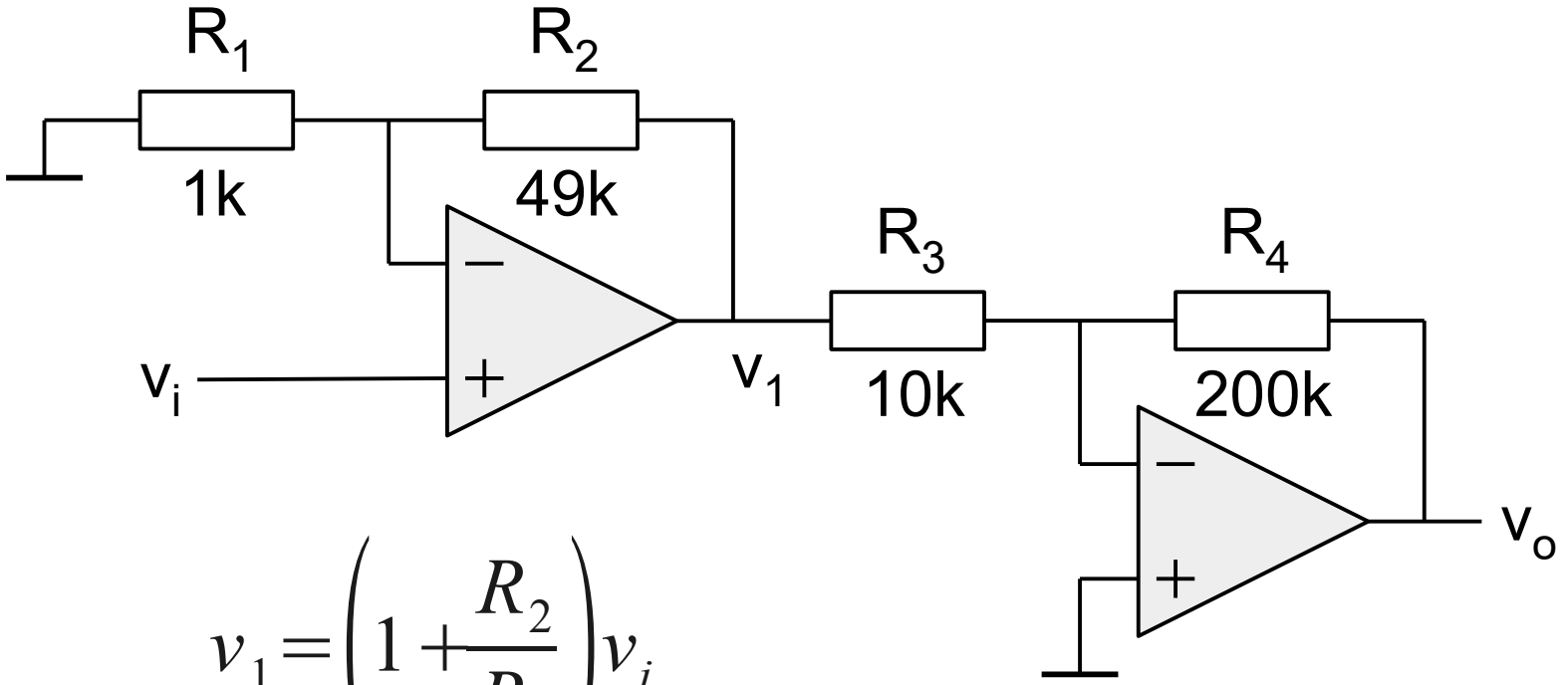
cu neidealități
și valori alese a.î.

$$R_3 \approx R_1' \| R_2$$

$$R_1' = R_1 + R_g, \quad R_g \ll R_1$$

$$v_o = -\frac{R_2}{R_1'} v_g + \left(1 + \frac{R_2}{R_1'}\right) V_{OS} + I_{OS} R_2$$

Exemplu: amplif. x1000 inversor

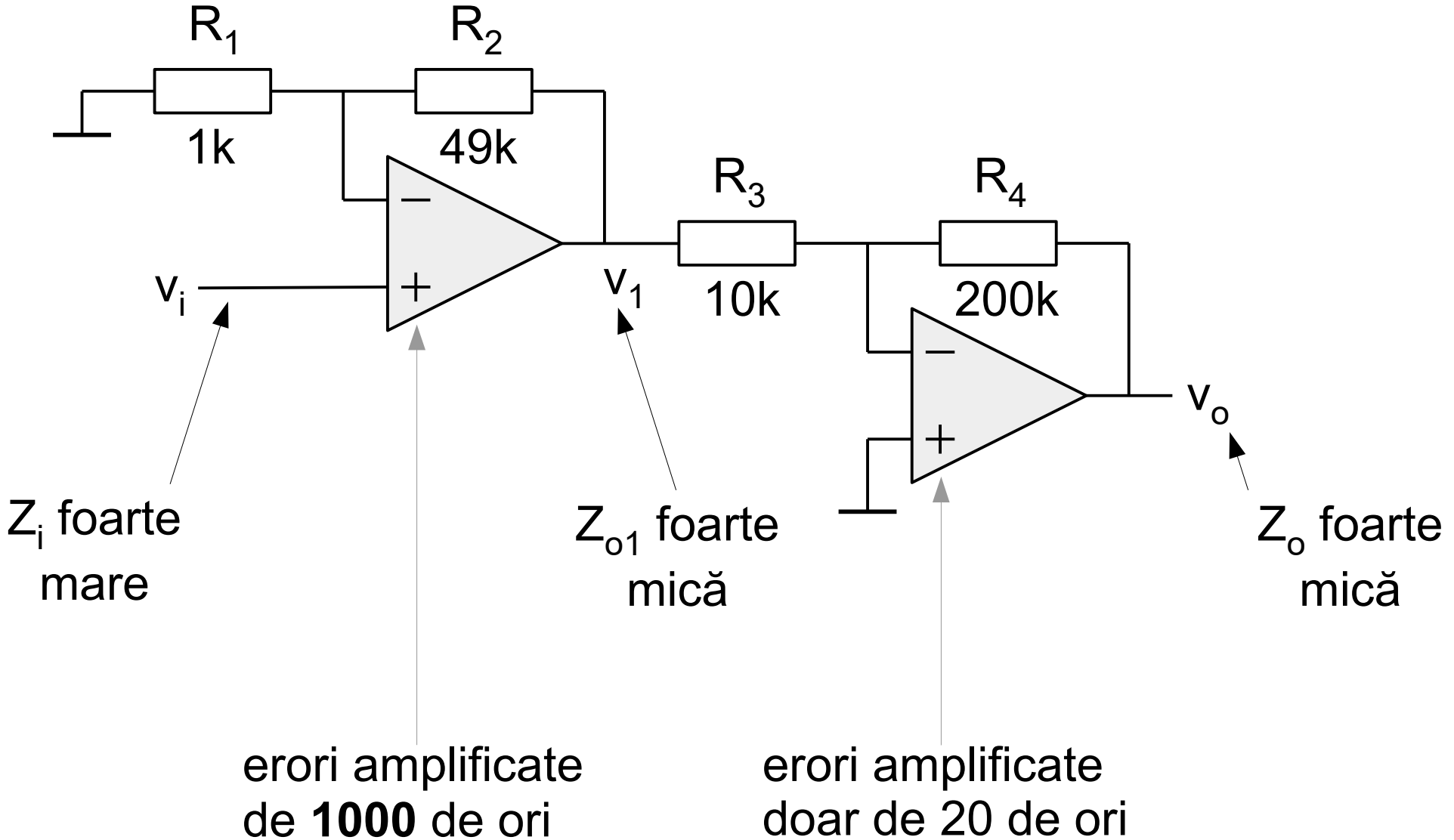


$$v_1 = \left(1 + \frac{R_2}{R_1}\right) v_i$$

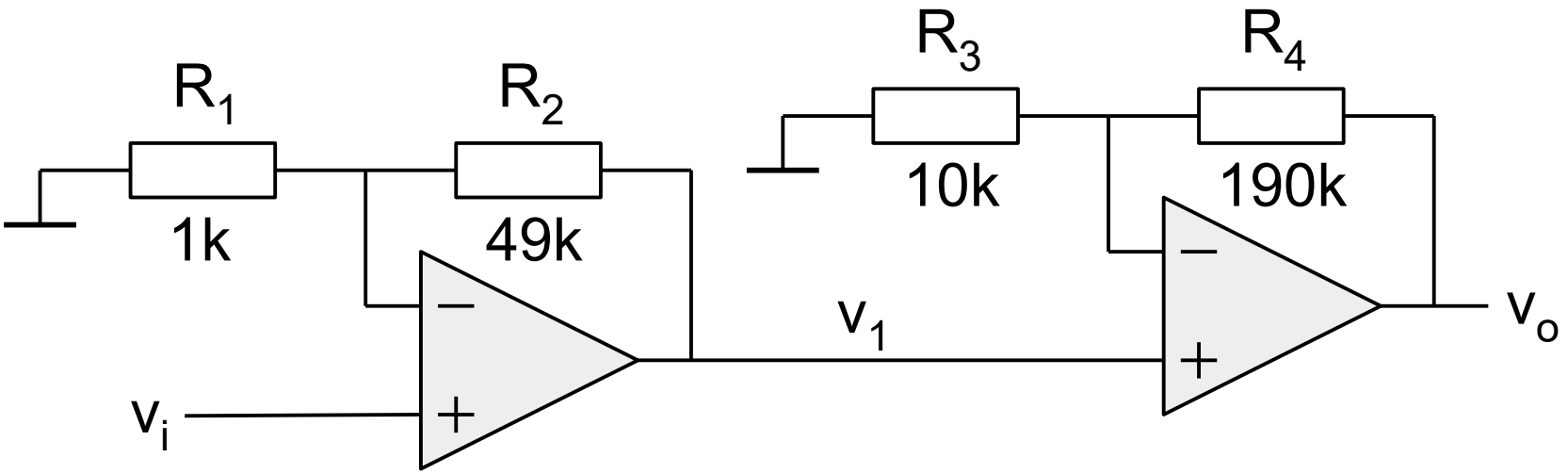
$$v_o = -\frac{R_4}{R_3} v_1$$

$$v_o = \left(1 + \frac{R_2}{R_1}\right)\left(-\frac{R_4}{R_3}\right) v_i ; \quad A_v = 50 \cdot (-20) = -1000$$

Comentarii


R1
R2
1k
49k
R3
R4
10k
200k
vi
v1
vo
Zi foarte mare
Zo1 foarte mică
Zo foarte mică
erori amplificate de 1000 de ori
erori amplificate doar de 20 de ori

Exemplu: amplif. x1000 neinversor



$$v_1 = \left(1 + \frac{R_2}{R_1}\right) v_i \qquad v_o = \left(1 + \frac{R_4}{R_3}\right) v_1$$

$$v_o = \left(1 + \frac{R_2}{R_1}\right)\left(1 + \frac{R_4}{R_3}\right) v_i ; \quad A_v = 50 \cdot 20 = 1000$$

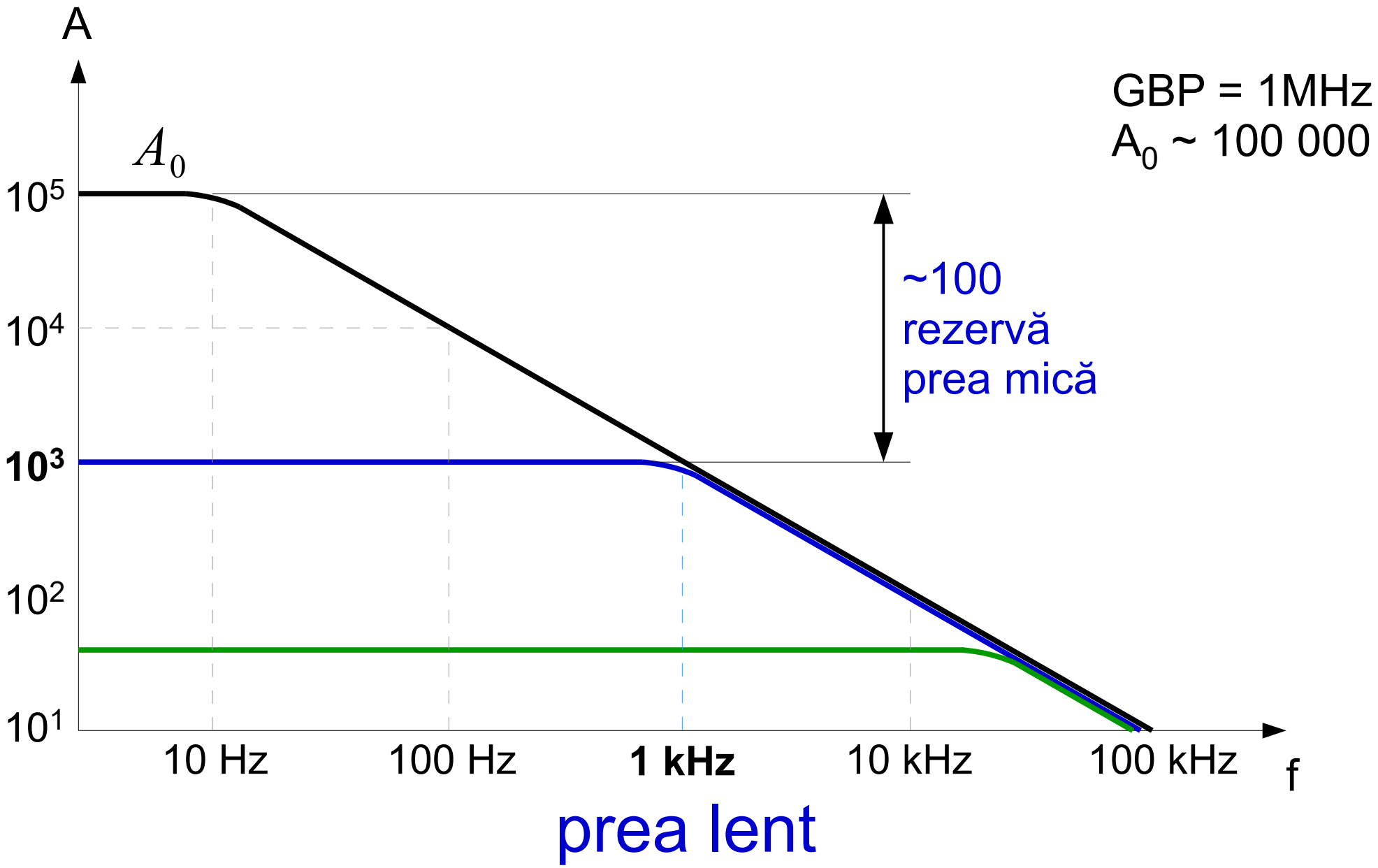
Comentarii (1)

- 49 kΩ, 190 kΩ **nu** sunt **valori standard**
  - trebuie găsite alte valori standard cu rapoarte corespunzătoare
  - *web search: resistor ratio calculator*
  - exemplu:

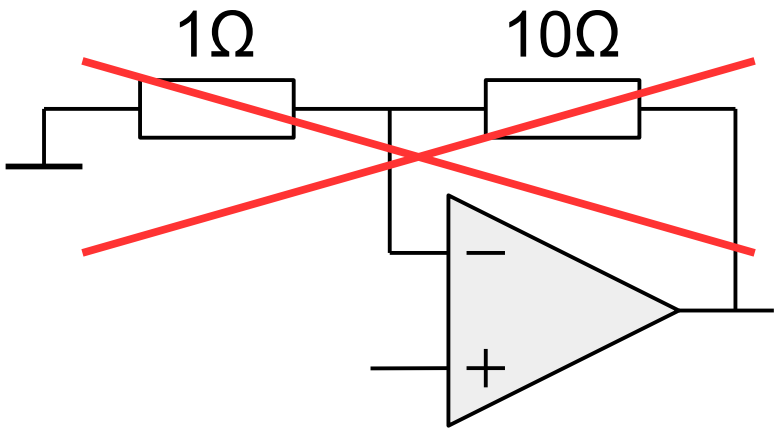
neinversor:	39k / 1k	=> 40
inversor:	100k / (1k8 + 2k2)	=> -25
total:		-1000

- de ce nu realizez montajul cu un singur AO?

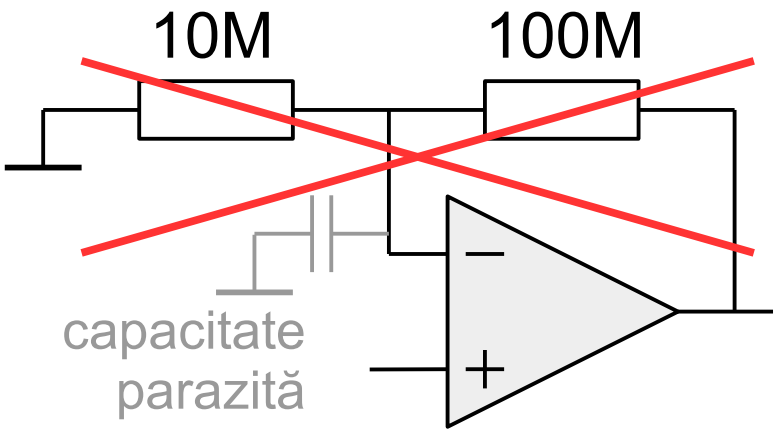
Comentarii (2)



Alegerea valorilor rezistențelor (1)

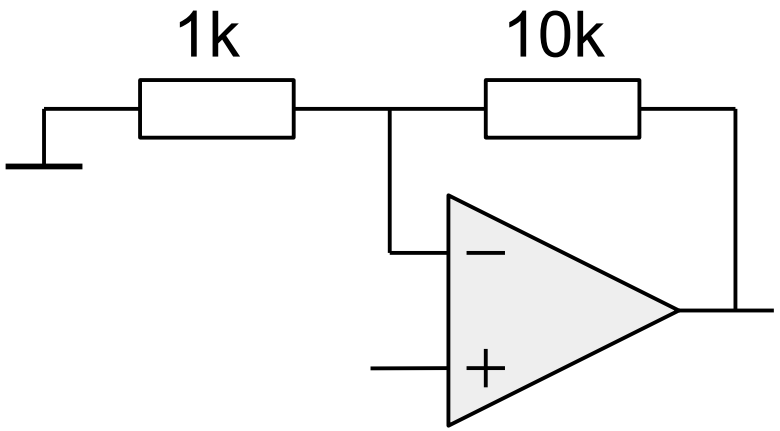


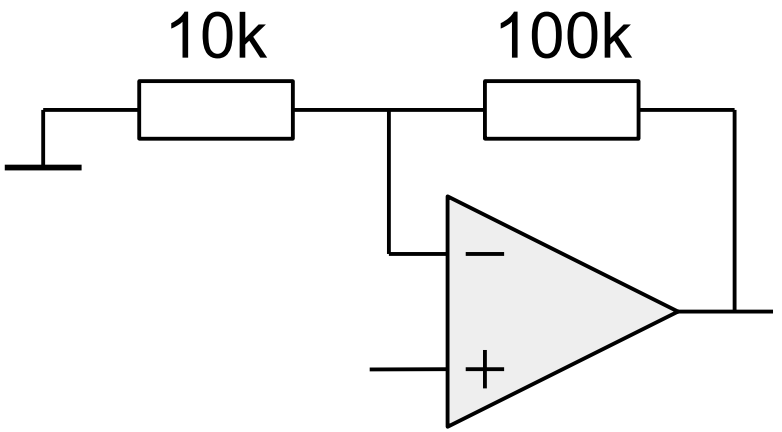
prea mici – AO intră în **limitare de curent** (protecție la scurt-circuit)



prea mari – efectul I_B, performanțe dinamice slabe

Valori rezonabile. Există valori **optime**?





Alegerea valorilor rezistențelor (2)

- Rezistențe mici
  - **zgomot mic**
  - consum mare
    încălzire => variază V_{OS}
    scădere a A_0
- Rezistențe mari
  - **consum mic**
  - efect sporit al I_B
  - zgomot mare

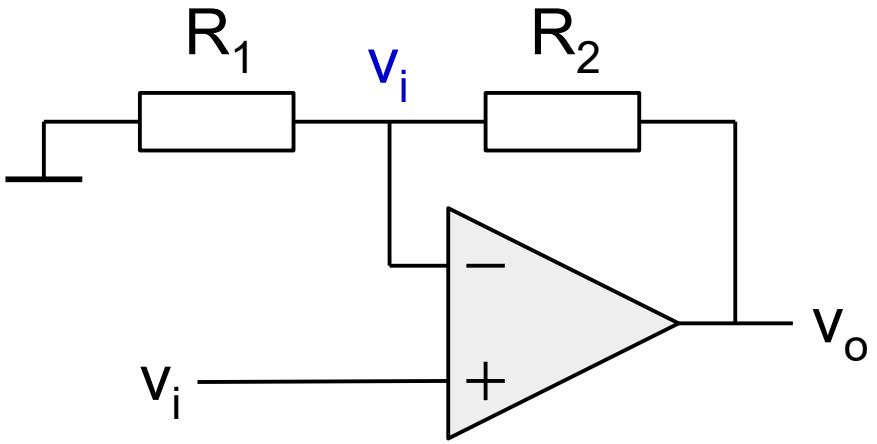
- Orice rezistență prezintă zgomot termic (*Johnson noise*)

$$P_n = k_B T \Delta f \qquad e_{n,R} = \sqrt{k_B T R \Delta f} \qquad 1\,\mathrm{k}\Omega: \approx 4\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$$

valoarea eficace a tensiunii de zgomot

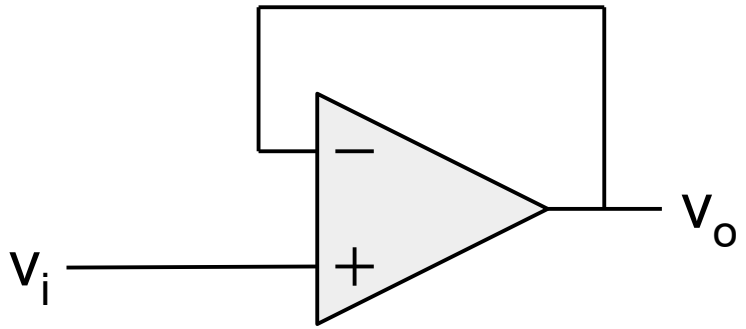
- Orice AO prezintă pe intrare:
  zgomot de tensiune (e_n) – fluctuație aleatoare a V_{OS}
  zgomot de curent (i_n) – fluctuație aleatoare a I_B

Repetorul (1)



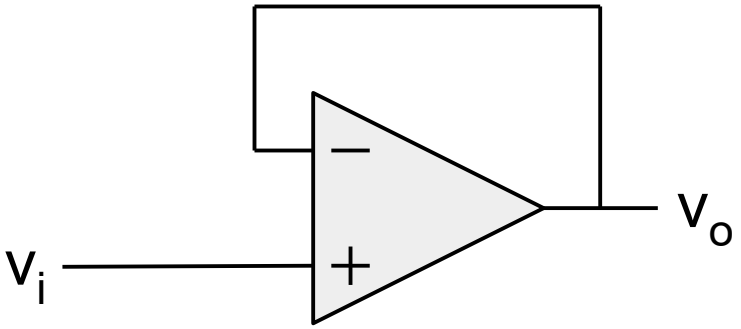
$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_i$$

$R_2 \to 0$ sau $R_1 \to \infty$: $v_o \to v_i$

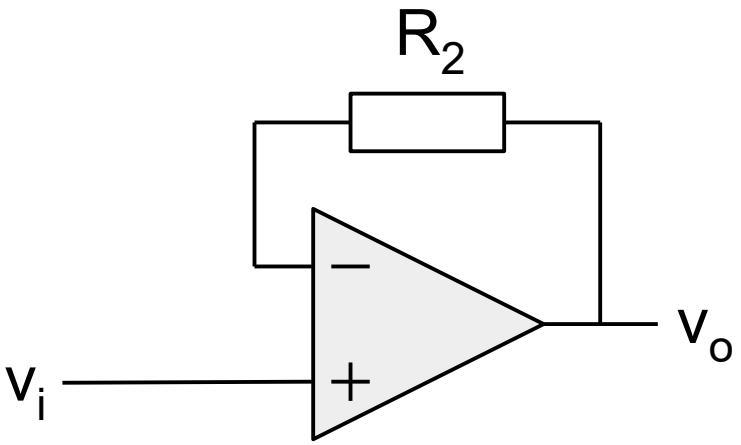


$$v_o = v_i$$

Repetorul (2)



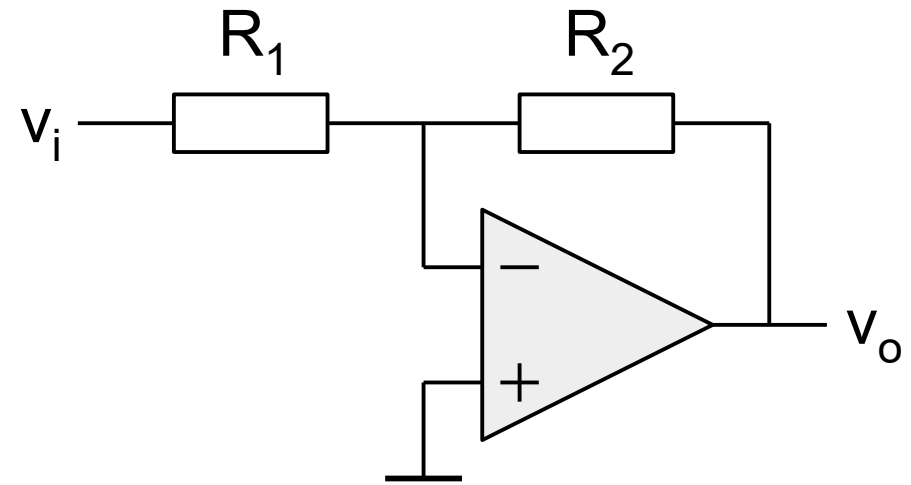
util când nu avem nevoie să amplificăm tensiunea, doar să realizăm **adaptarea de impedanță**: Z_i foarte mare, Z_o foarte mic



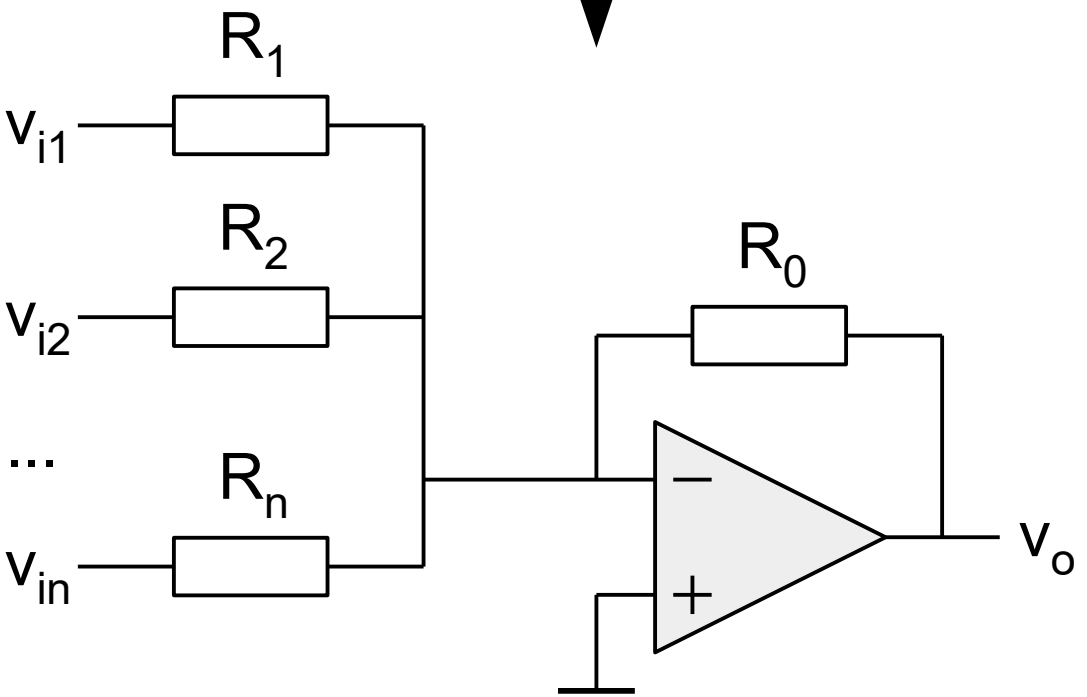
variantă cu compensarea curentului de polarizare

$$R_2 \approx R_g$$

Sumatorul (1)

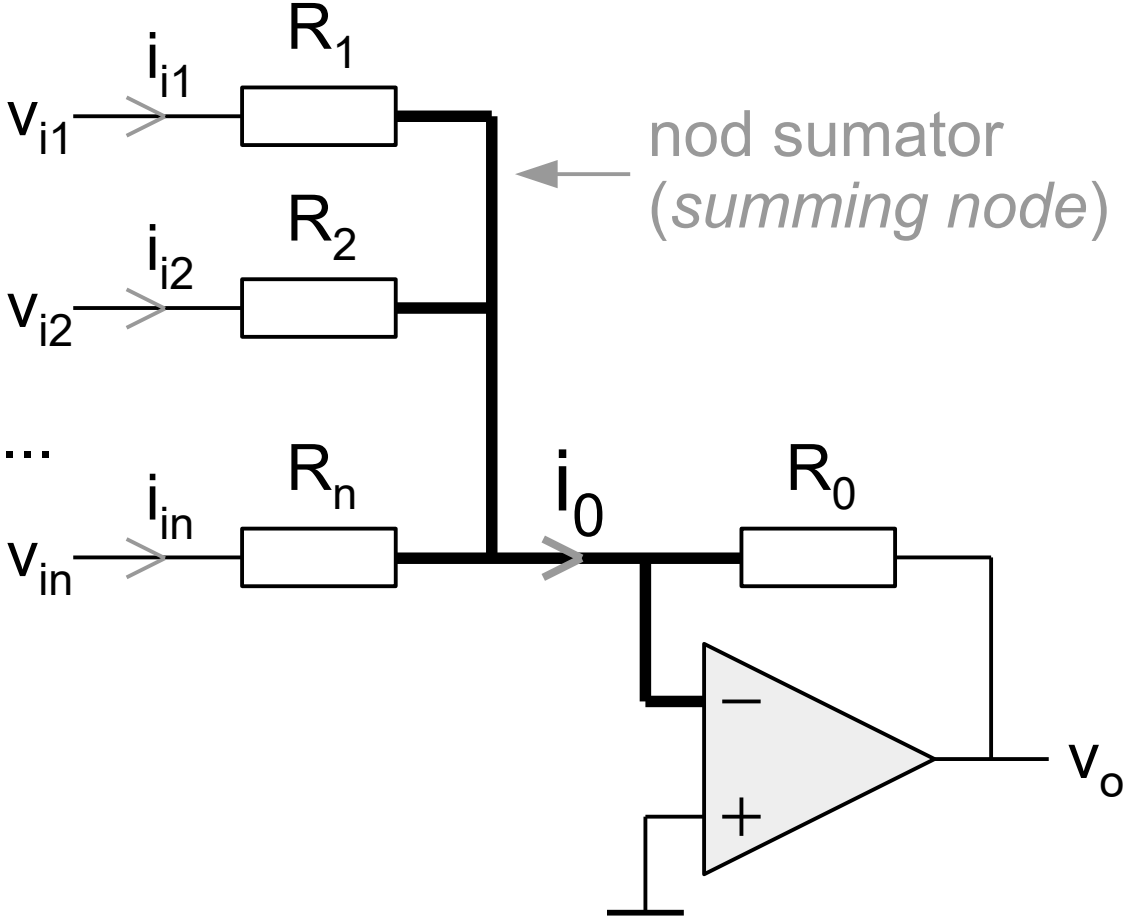


$$v_o = -\frac{R_2}{R_1} v_i$$



$$v_o = -\sum_{k=1}^{n} \frac{R_0}{R_k} v_{ik}$$

Sumatorul (2)



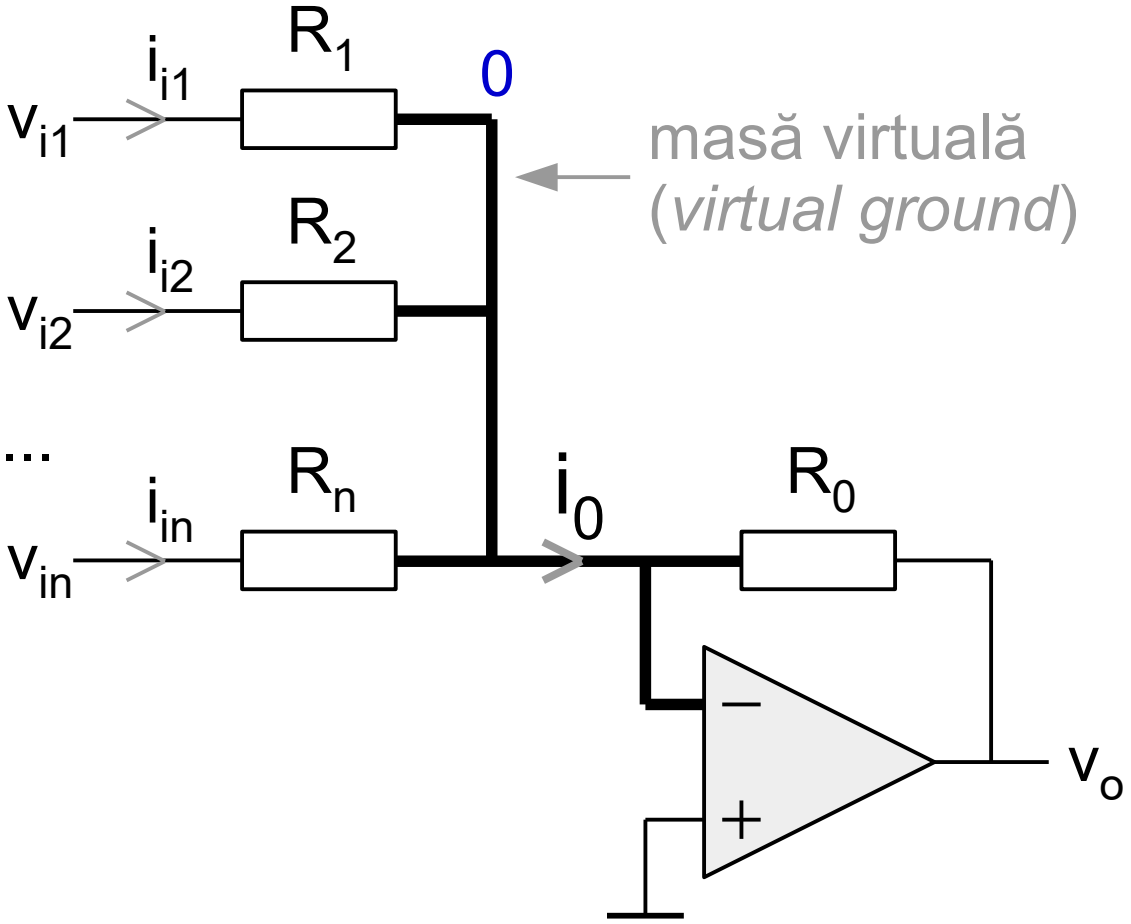
$$i_0 = \sum_{k=1}^{n} i_{ik}$$

$$i_{ik} = \frac{v_{ik}}{R_k}$$

$$v_o = -i_0 R_0$$

$$v_o = -\sum_{k=1}^{n} \frac{R_0}{R_k} v_{ik}$$

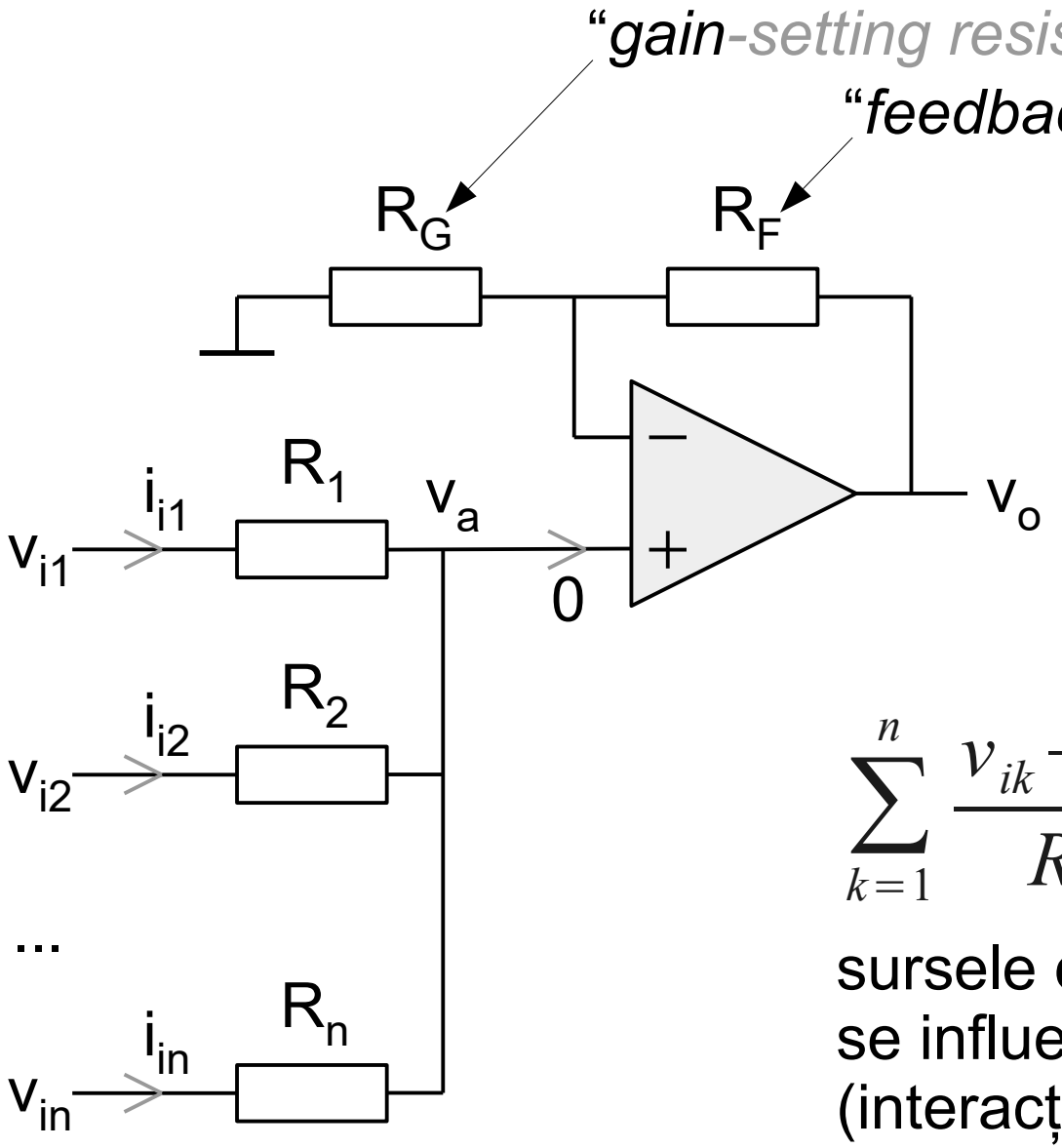
Sumatorul (3)



$$i_{ik} = \frac{v_{ik}}{R_k} = \frac{v_{gk}}{R_k + R_{gk}}$$

sursele de intrare
nu se influențează reciproc
(nu comunică)
(nu interacționează)

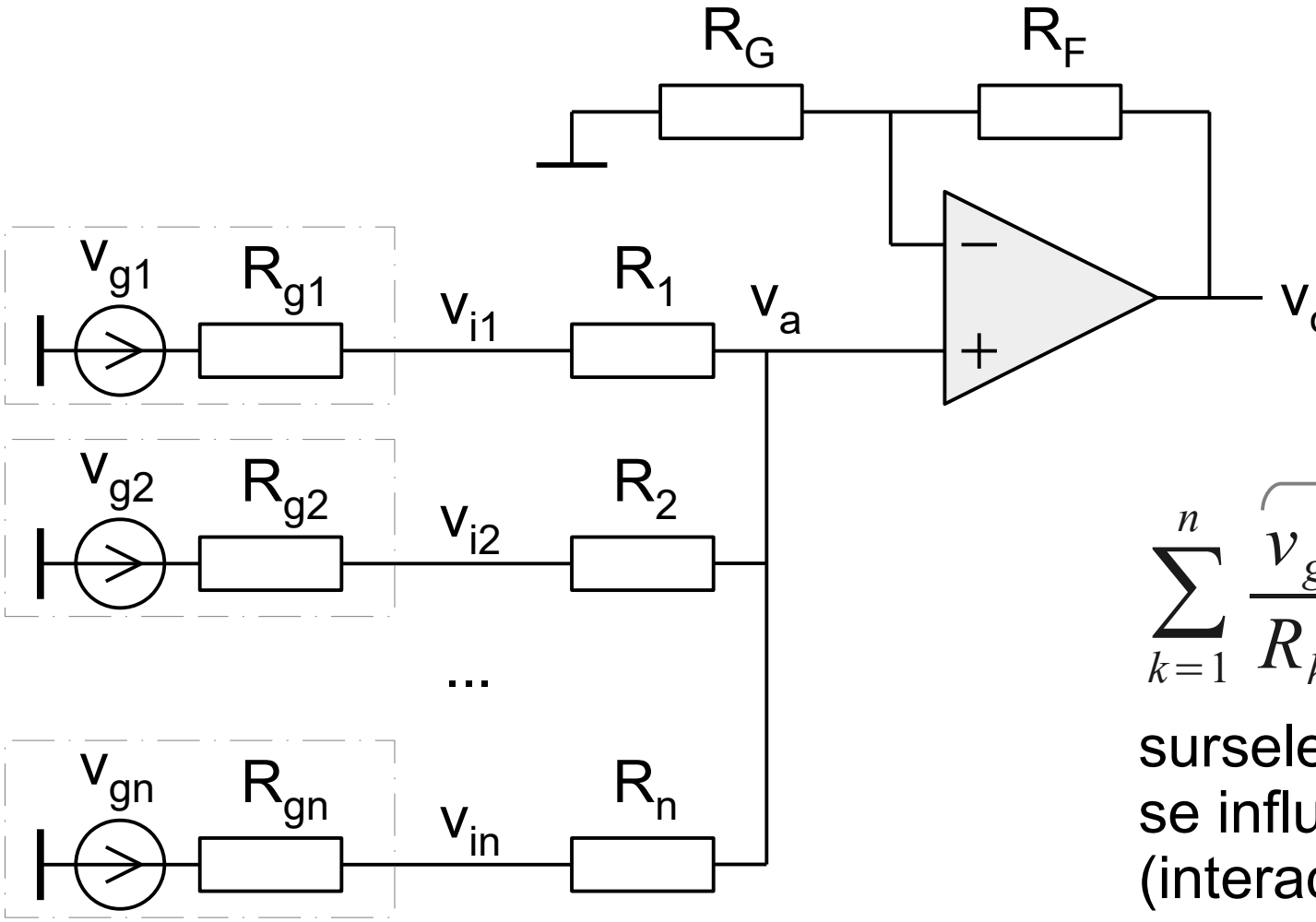
Sumatorul neinversor (1)



$$\sum_{k=1}^{n} \frac{v_{ik} - v_a}{R_k} = 0$$

sursele de intrare
se influențează reciproc
(interacționează)

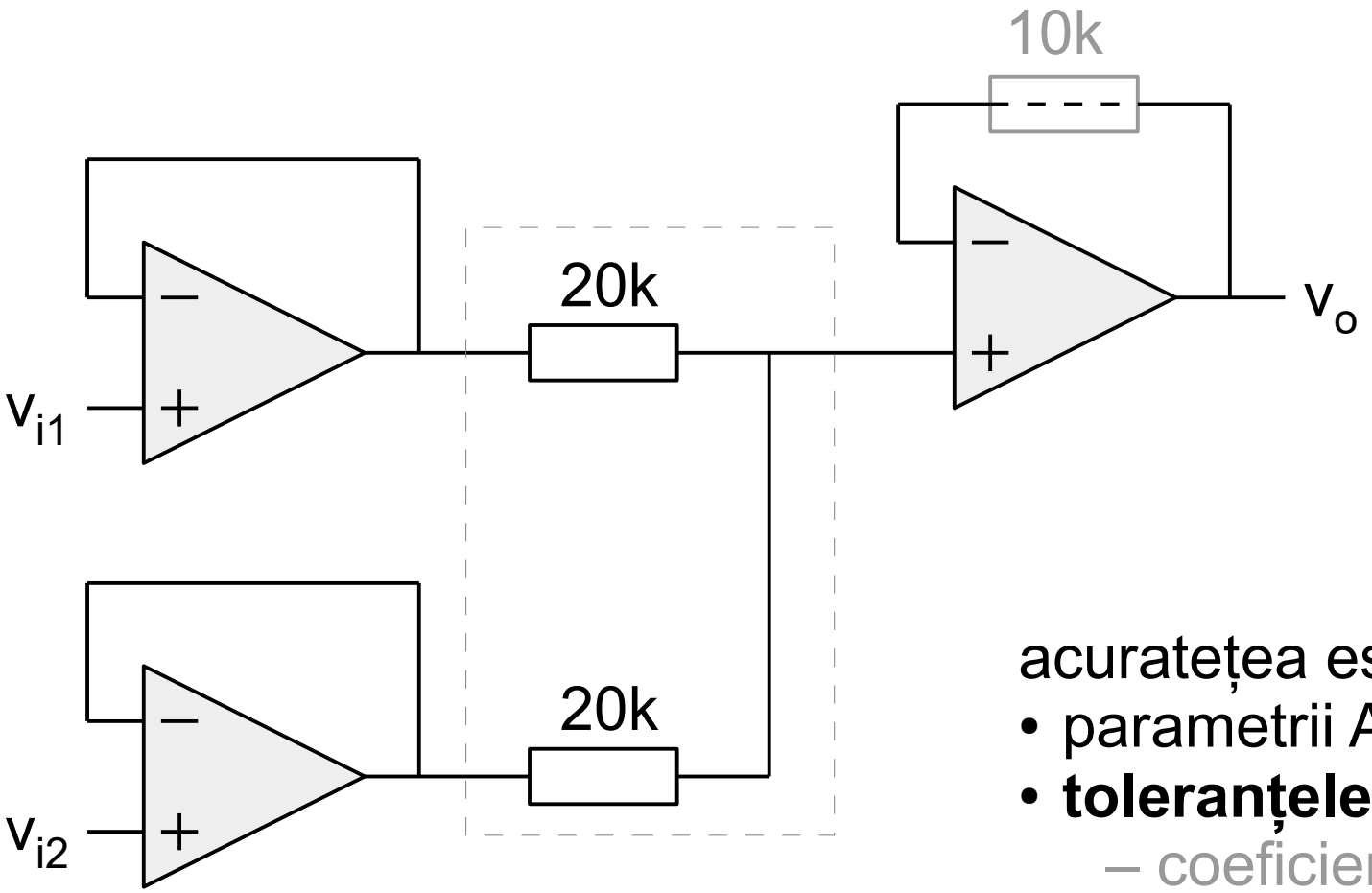
Sumatorul neinversor (2)



$$\sum_{k=1}^{n} \frac{\overbrace{v_{gk} - v_a}^{i_{ik}}}{R_k + R_{gk}} = 0$$

sursele de intrare
se influențează reciproc
(interacționează)

Exemplu: media a două semnale

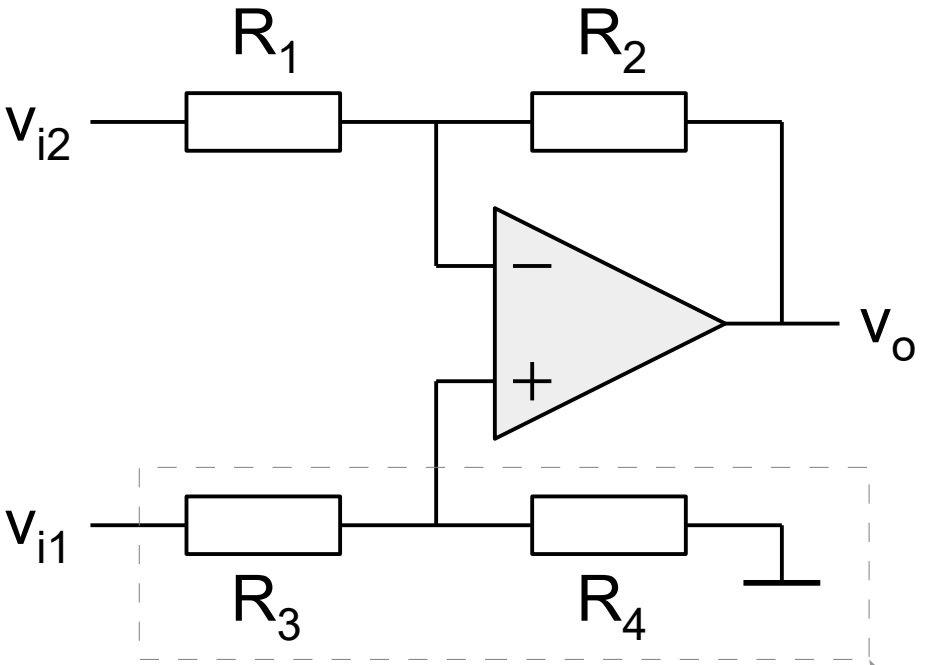


acuratețea este limitată de:

- parametrii AO
- **toleranțele rezistoarelor**
  - coeficienții termici
  - îmbătrânirea

Există rezistoare “împerecheate” fabricate sub formă monolitică (de circuit integrat).

Amplificatoare de diferență (1)



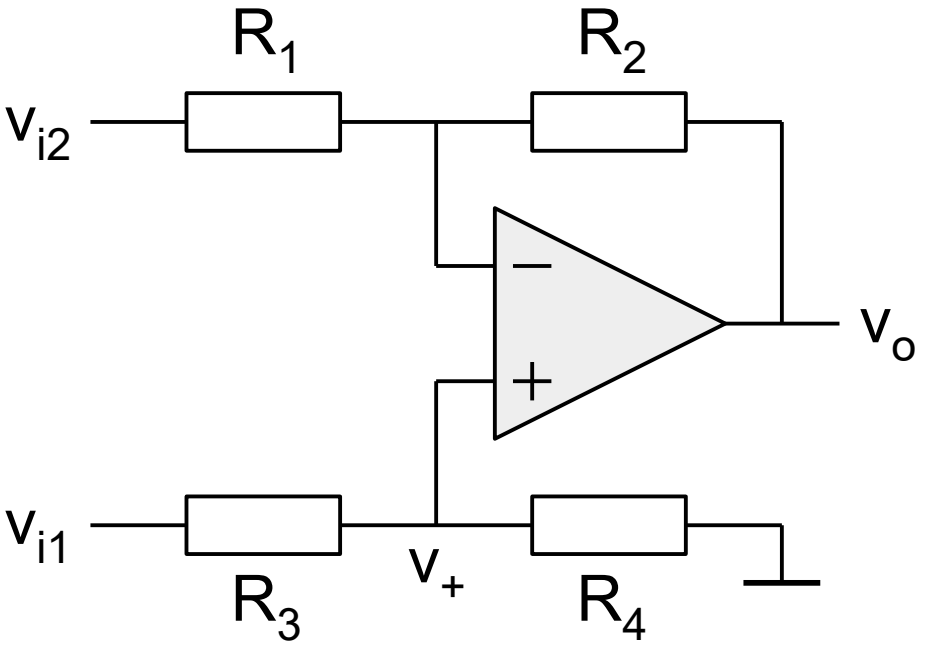
problemă:
a amplifica diferența dintre tensiunile de intrare

idee:
superpoziție inversor+neinversor

dar:
calea neinversoare are amplificarea mai mare decât cea inversoare

soluție:
introducem un divizor de tensiune care să compenseze acest efect

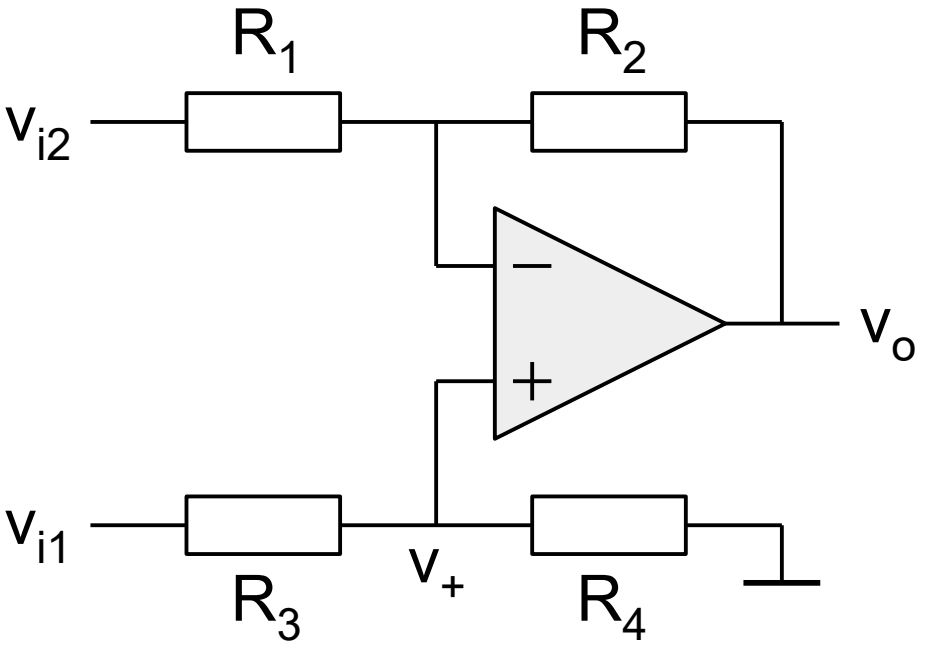
Amplificatoare de diferență (2)



$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_+ - \frac{R_2}{R_1} v_{i2}$$

$$v_+ = v_{i1} \frac{R_4}{R_3 + R_4} \longrightarrow v_o = \left(1 + \frac{R_2}{R_1}\right) \frac{R_4}{R_3 + R_4} v_{i1} - \frac{R_2}{R_1} v_{i2}$$

Amplificatoare de diferență (3)



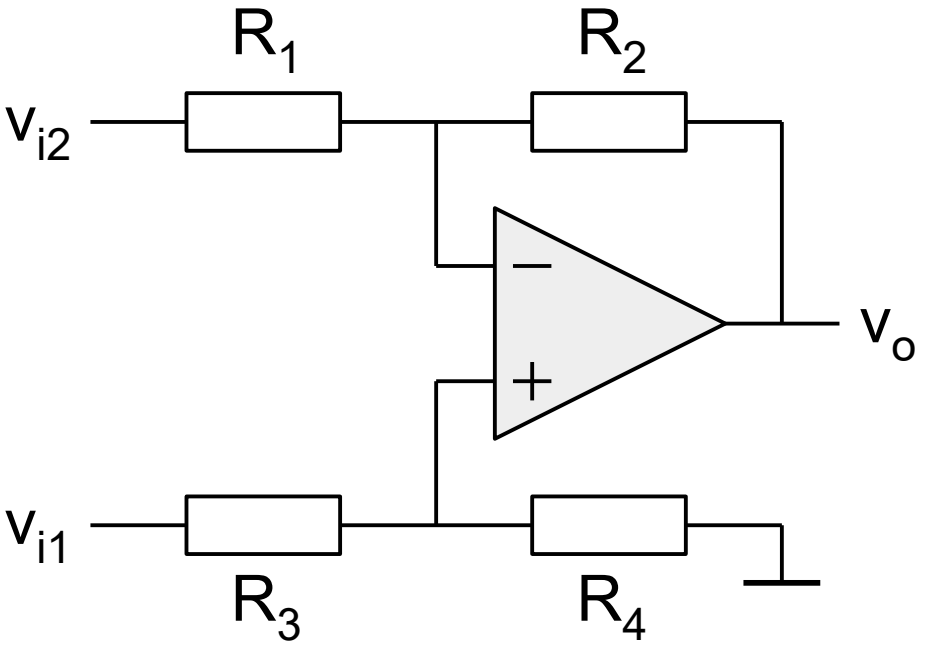
$$v_o = \left(1 + \frac{R_2}{R_1}\right)\frac{R_4}{R_3 + R_4} v_{i1} - \frac{R_2}{R_1} v_{i2}$$

condiția de amplificator diferențial:

$$\left(1 + \frac{R_2}{R_1}\right)\frac{R_4}{R_3 + R_4} = \frac{R_2}{R_1}$$

atunci: $$v_o = \frac{R_2}{R_1}\left(v_{i1} - v_{i2}\right)$$

Amplificatoare de diferență (4)



condiția de amplificator diferențial:

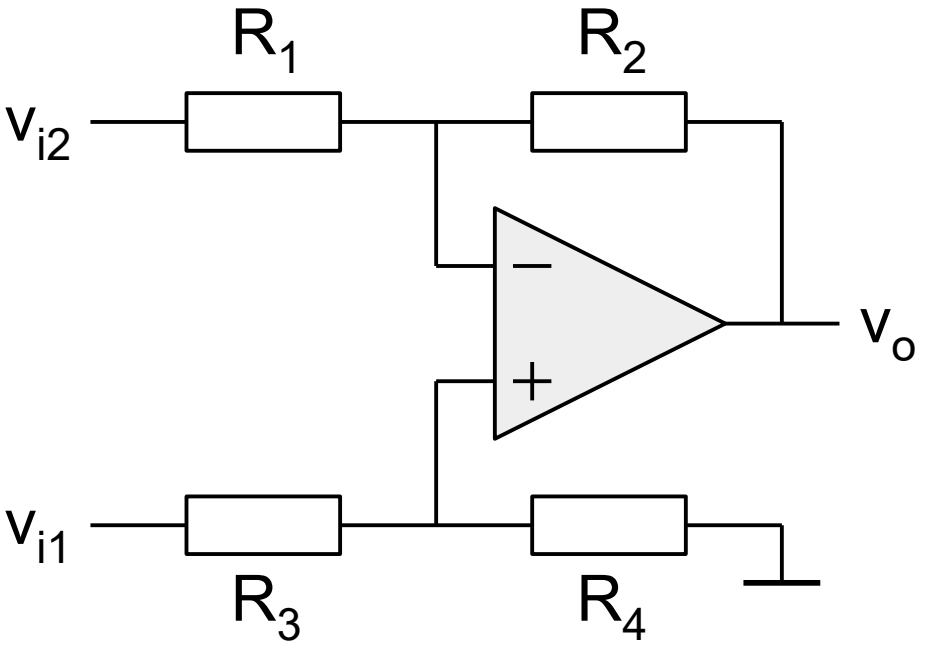
$$\left(1+\frac{R_2}{R_1}\right)\frac{R_4}{R_3+R_4}=\frac{R_2}{R_1}$$

elaborând:

$$\frac{R_4}{R_3+R_4}=\frac{R_2}{R_1+R_2}$$

$$\boxed{\frac{R_4}{R_3}=\frac{R_2}{R_1}}$$

Amplificatoare de diferență (5)



$$\frac{R_4}{R_3} = \frac{R_2}{R_1}$$

în general:

$$R_3 = R_1 \text{ și } R_4 = R_2$$

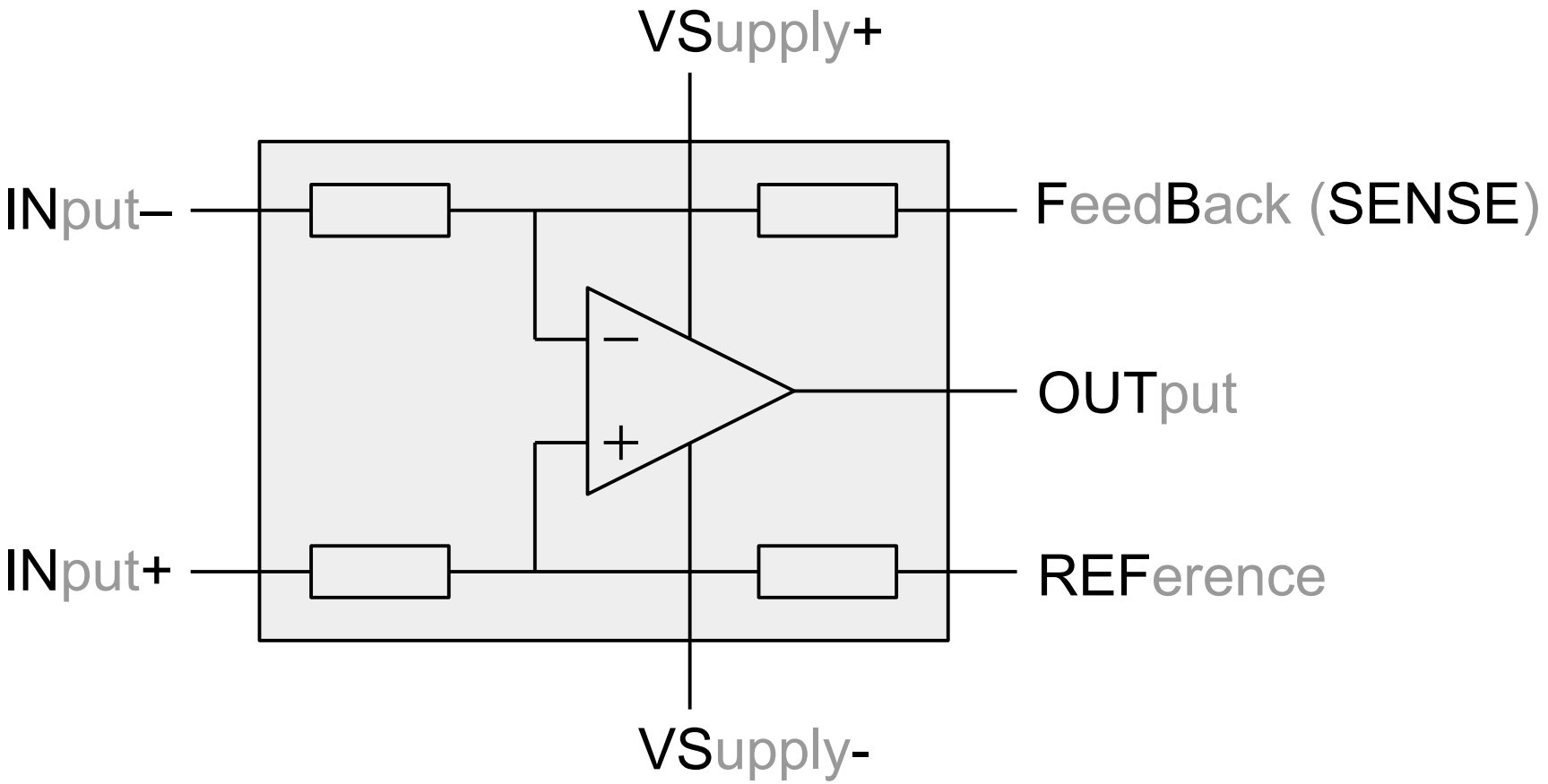
Funcționarea corespunzătoare (asigurarea rejecției modului comun) se bazează pe respectarea acestei egalități.

Realizarea cu rezistoare discrete duce la performanțe slabe.
Se preferă piese fabricate monolitic (circuit integrat: AO + 4 rezistențe)

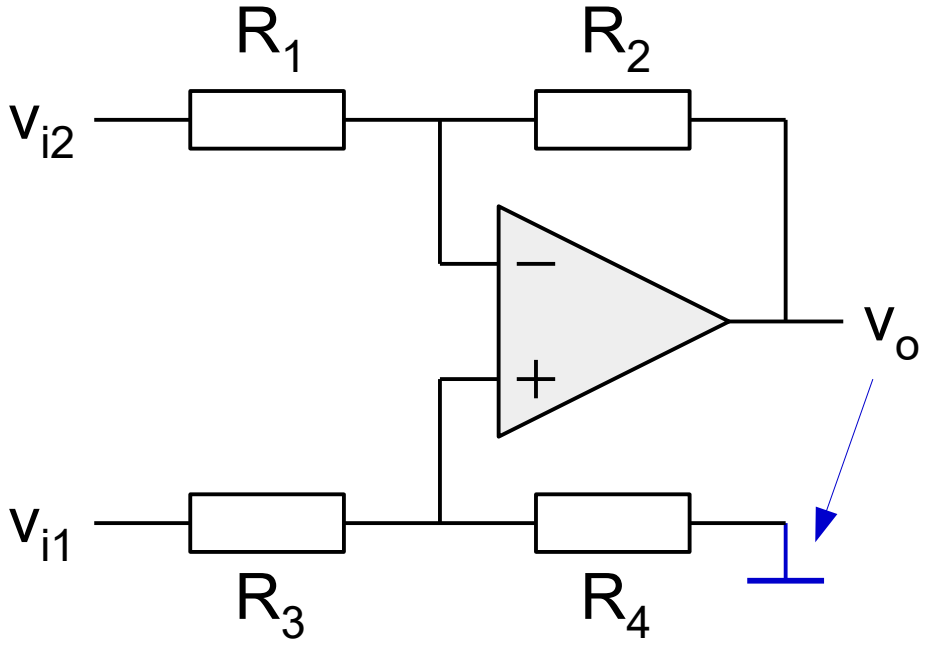
Intrările trebuie să provină de la surse cu impedanță foarte mică (AO).

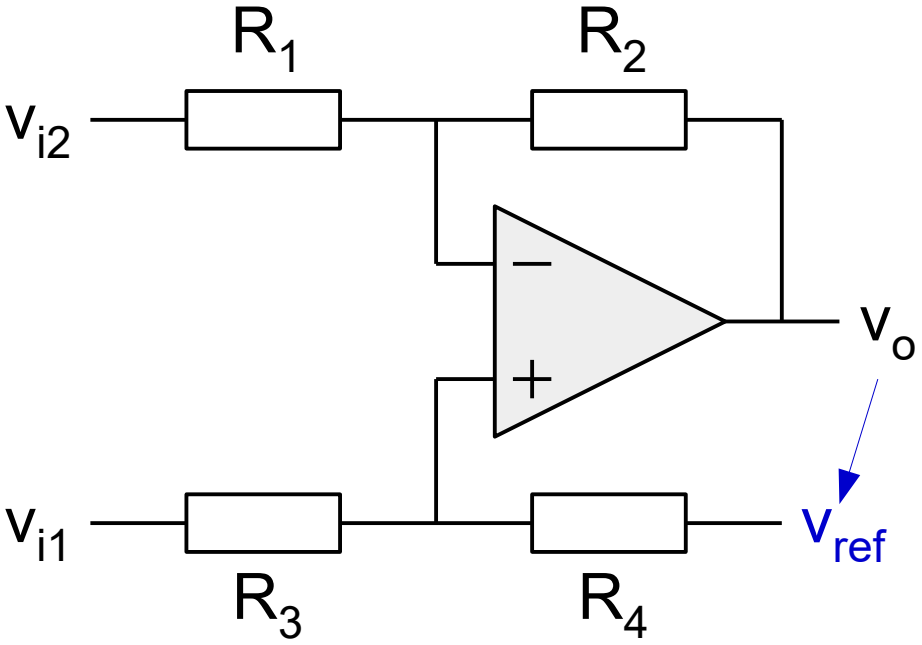
AD – realizare integrată

Realizare sub formă de circuit integrat (“*difference amplifier*”):



AD – masă / a treia intrare

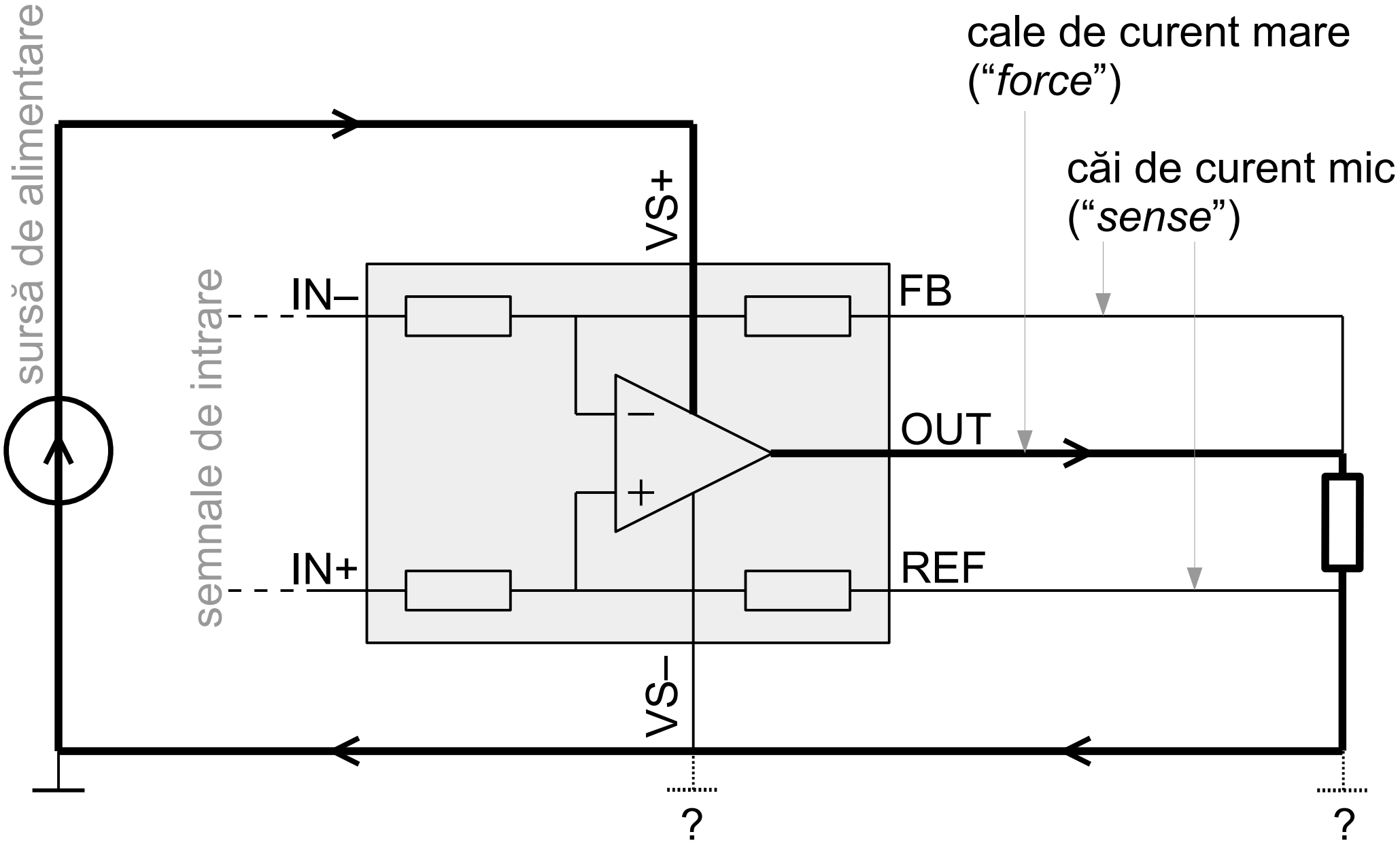




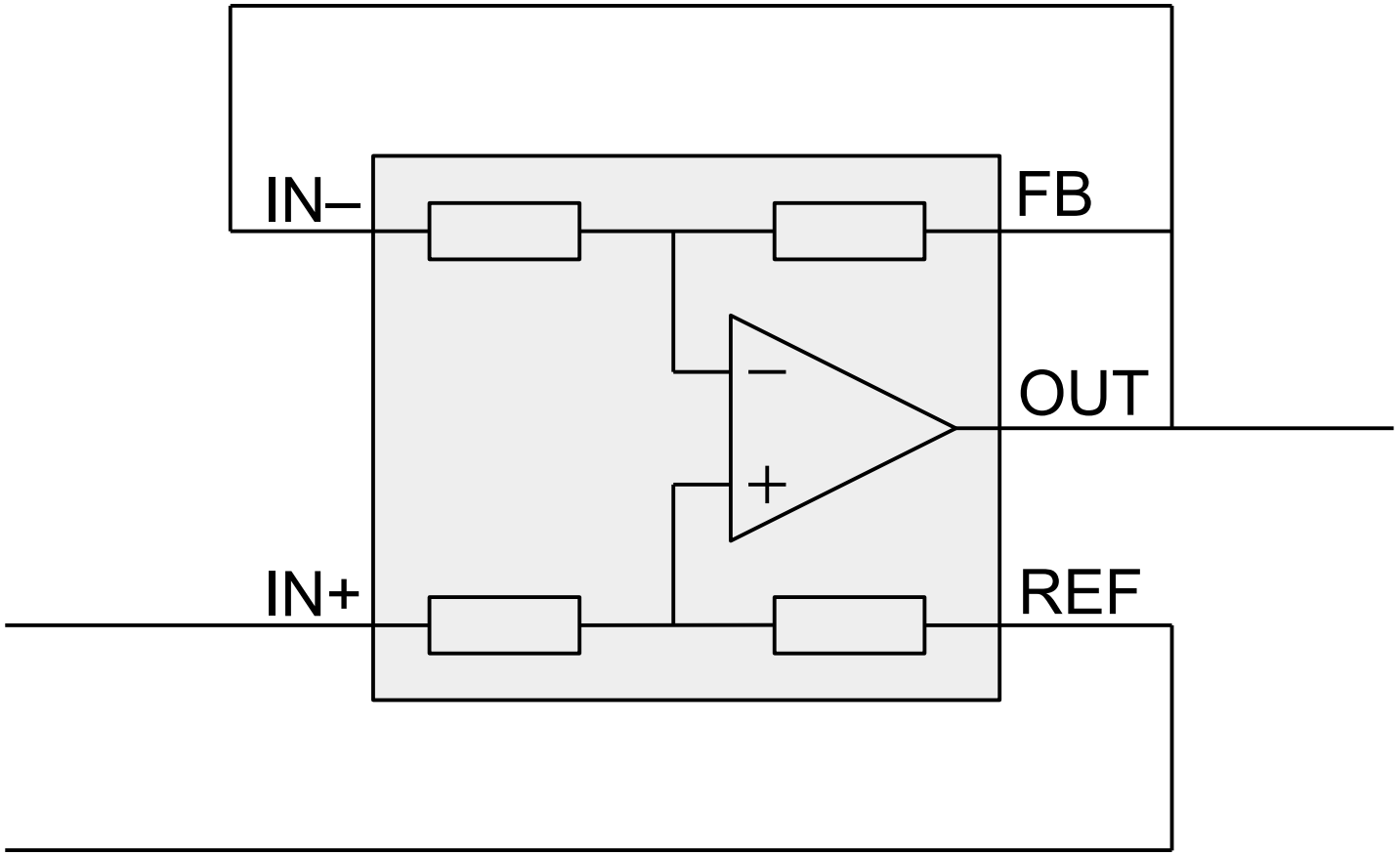
$$v_o = \frac{R_2}{R_1}\left(v_{i1} - v_{i2}\right)$$

$$v_o = \frac{R_2}{R_1}\left(v_{i1} - v_{i2}\right) + v_{ref}$$

AD – “*Remote sensing*”

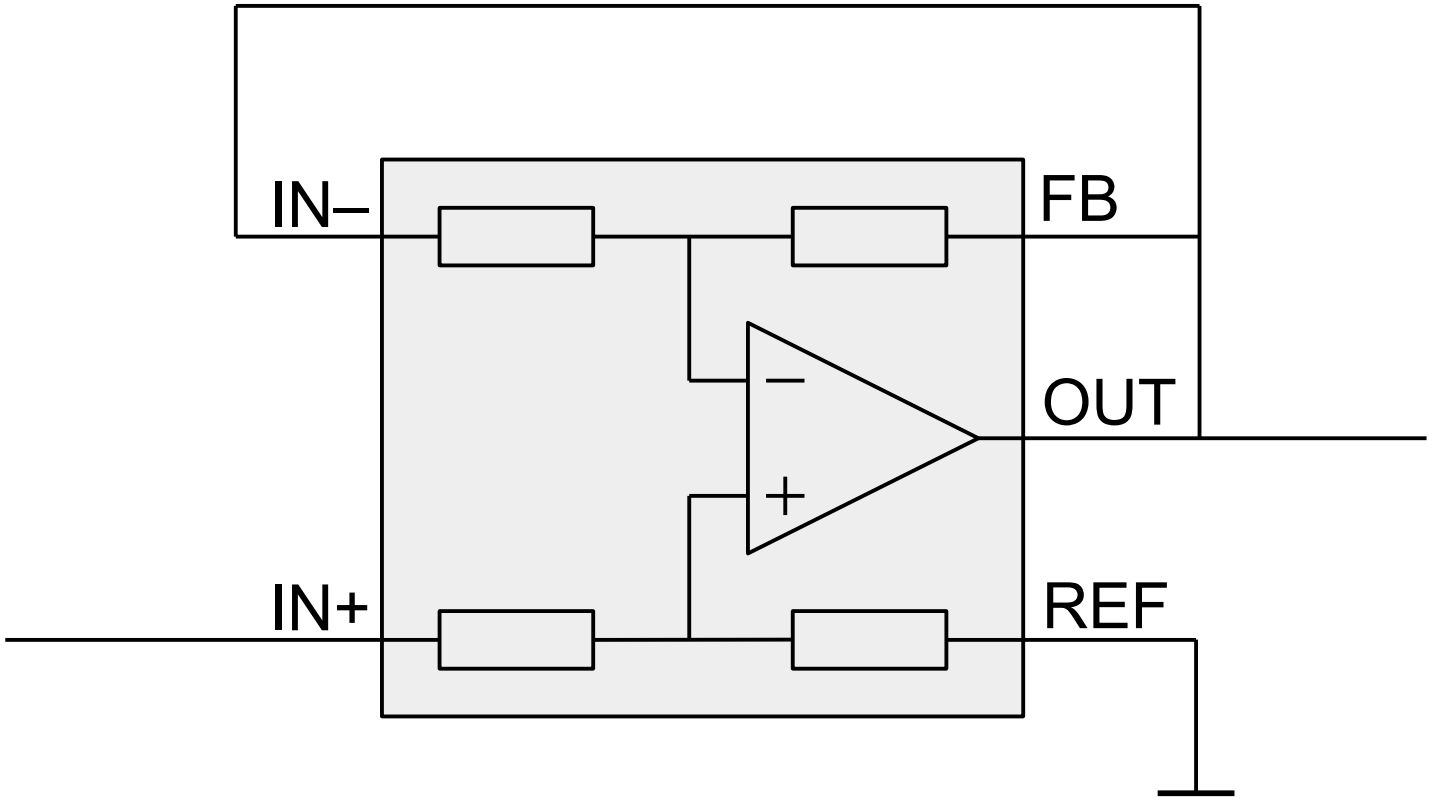


AD integrat – Alte utilizări (1)



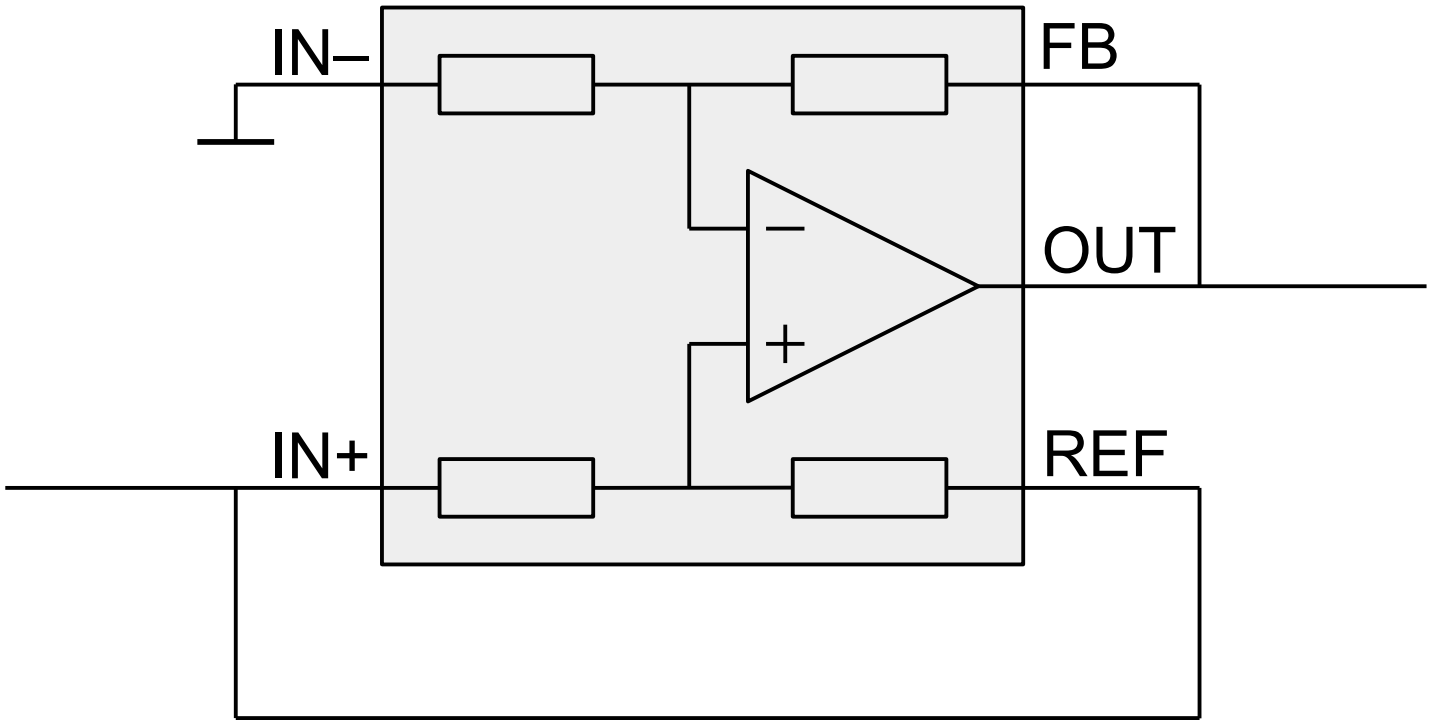
Realizarea mediei **cu precizie**

AD integrat – Alte utilizări (2)



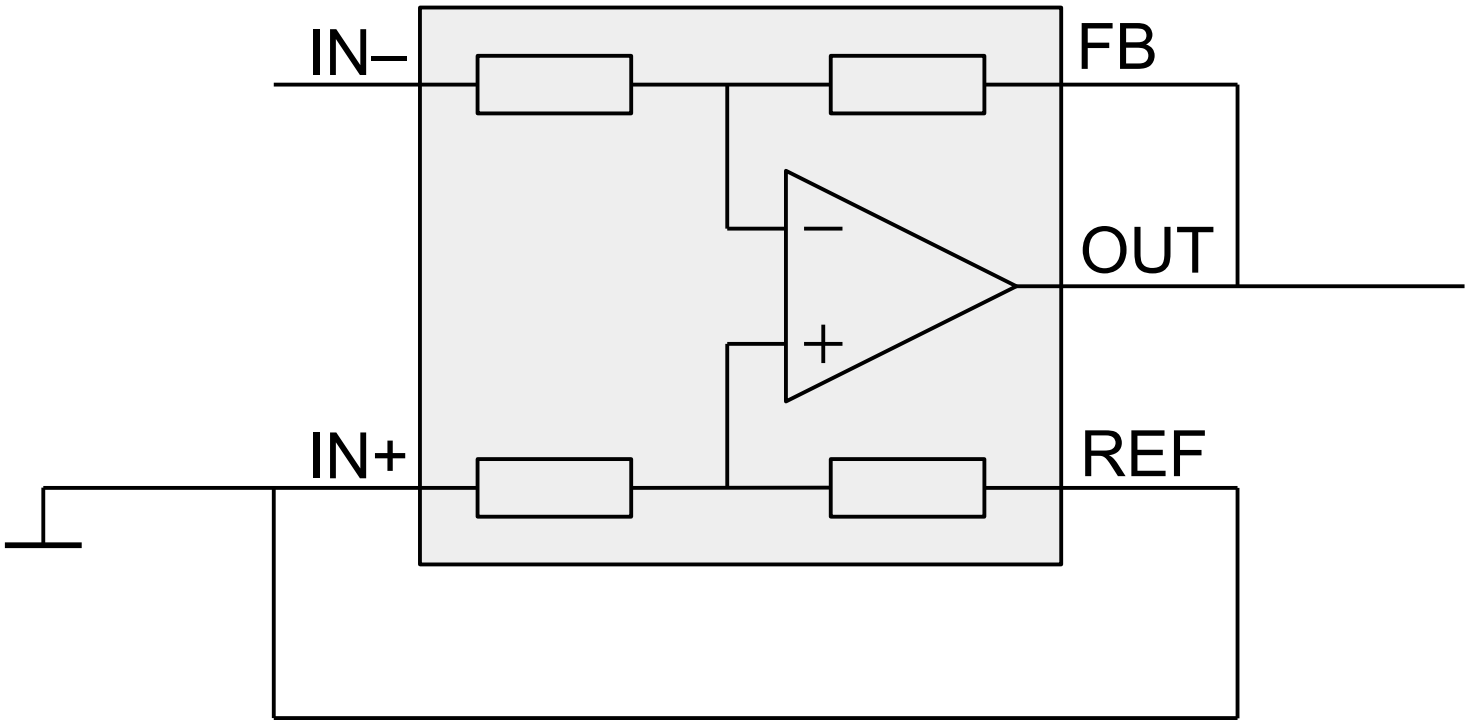
Divizor de tensiune **de precizie**

AD integrat – Alte utilizări (3)



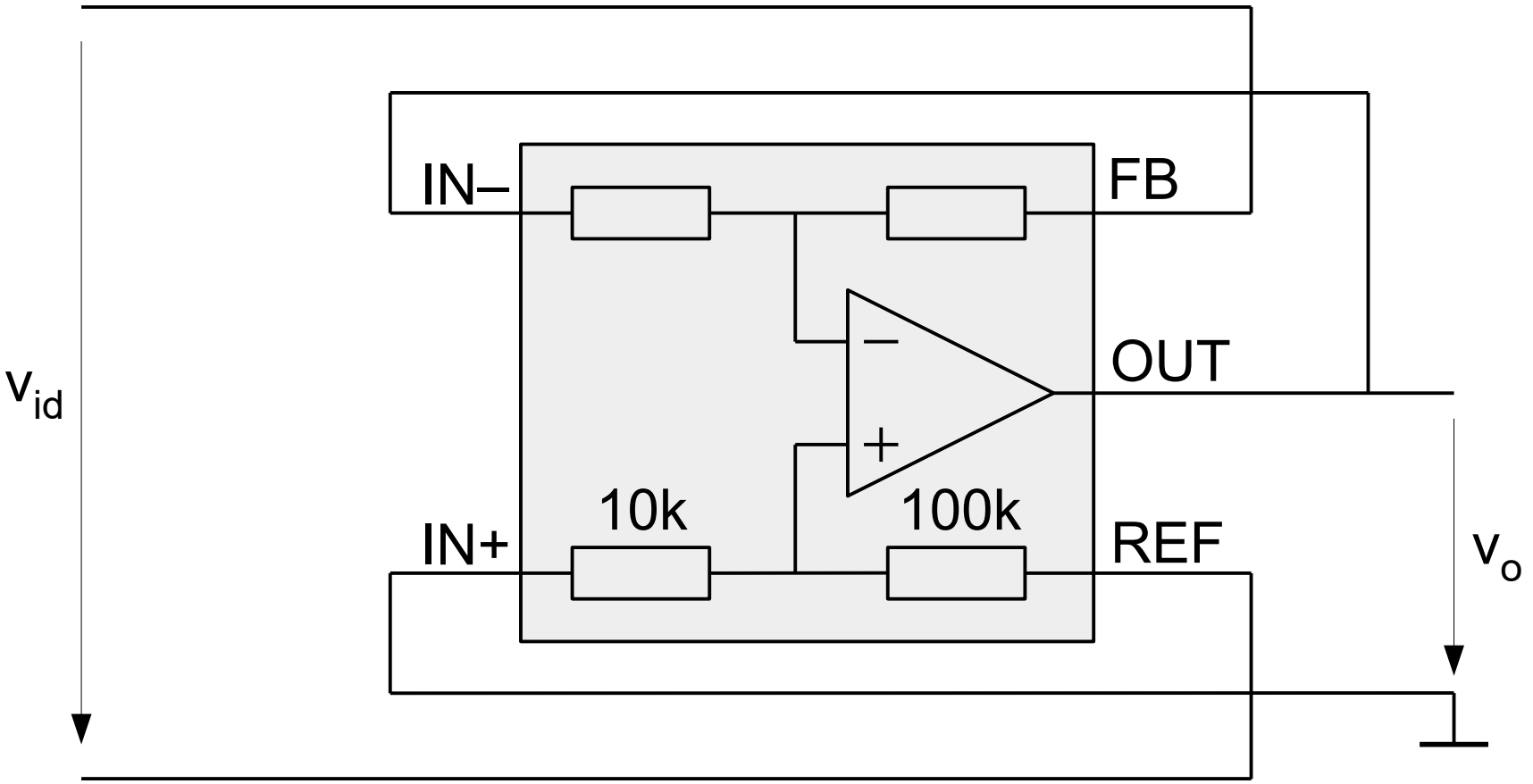
Amplificator neinversor **de precizie**

AD integrat – Alte utilizări (4)



Amplificator inversor **de precizie**

AD integrat – Alte utilizări (5)



Amplificator diferențial **cu domeniu larg de intrare**